

**MONITORING OF A PEDIATRIC LOWER LIMB
EXOSKELETON FOR PARAPLEGIA DURING
REHABILITATION**

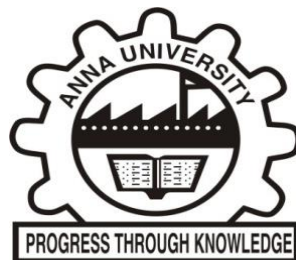
PROJECT WORK II

Submitted by

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in partial fulfilment of the requirement for the degree of

**MASTER OF ENGINEERING IN
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**DEPARTMENT OF PRODUCTION TECHNOLOGY
MADRAS INSTITUTE OF TECHNOLOGY CAMPUS
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ANNA UNIVERSITY: CHENNAI 600 044**BONAFIDE CERTIFICATE**

Certified that this project report entitled "**MONITORING OF A PEDIATRIC LOWER LIMB EXOSKELETON FOR PARAPLEGIA DURING REHABILITATION** " is the bona-fide work of **AADHI JEGAN NARAYANAN K N (2024608031)** who carried out the work under my supervision .Certified further that, to the best of my knowledge, the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate

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ABSTRACT

This work presents the experimental implementation, control, and analysis of a pediatric lower-limb exoskeleton prototype developed for slow and controlled gait rehabilitation applications. The system employs a potentiometer-based closed-loop control architecture to achieve coordinated bilateral hip and knee joint motion using DC motors, BTS7960 motor drivers, and an Arduino Mega controller. Realistic gait trajectories were generated for both legs with alternate phase coordination, enabling stationary walking motion that closely resembles natural human gait patterns. The experimentally obtained joint trajectories achieved hip flexion–extension motion within -25° to $+40^\circ$ and knee motion within -30° to $+45^\circ$, corresponding to safe pediatric gait ranges.

Angular velocity and acceleration analyses demonstrated smooth and stable gait dynamics suitable for low-speed rehabilitation. The hip joint exhibited angular velocity within approximately ± 20 – $50^\circ/\text{s}$ and angular acceleration within ± 50 – $120^\circ/\text{s}^2$, while the knee joint showed slightly higher dynamic behavior with velocity ranging from ± 25 – $60^\circ/\text{s}$ and acceleration reaching ± 60 – $150^\circ/\text{s}^2$. The generated waveforms were periodic, continuous, and free from abrupt transitions, confirming stable bilateral coordination and realistic gait sequencing. Error convergence analysis further demonstrated effective closed-loop operation, with tracking error gradually reducing toward the desired trajectory under PD-based control.

Torque analysis indicated that the hip joint requires higher actuator effort due to its primary load-supporting role, with torque demand maintained within approximately 20 Nm to +25 Nm, whereas the knee joint required comparatively lower torque within 8 Nm to +15 Nm. PWM control effort analysis demonstrated smooth actuator drive variation throughout the gait cycle, confirming efficient motor utilization without abrupt saturation or instability. The obtained control effort profiles validated the suitability of low-cost DC motor actuation for slow gait assistance applications.

The developed prototype also incorporated bilateral gait symmetry analysis, hip–knee coordination cyclograms, stance–swing phase segmentation, and real-time gait synchronization to validate physiologically realistic lower-limb motion. The implemented PD control strategy successfully maintained stable closed-loop tracking with steady-state error limited to approximately $\pm 2^\circ$ to $\pm 3^\circ$, despite practical limitations such as sensor noise, mechanical backlash, and friction effects.

Overall, the present work validates the feasibility of implementing a low-cost potentiometer-based bilateral exoskeleton gait control system for pediatric rehabilitation applications. The obtained experimental results demonstrate stable gait generation, smooth actuator response, realistic joint coordination, and reliable closed-loop performance, thereby providing a strong foundation for future development of intelligent intention-based exoskeleton systems incorporating EMG, IMU–encoder fusion, and EEG-assisted gait control.

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TABLE OF CONTENTS

CHAPTER No.	TITLE	PAGE No.
1	INTRODUCTION	1
	1.1 BACKGROUND	1
	1.2 NEED OF LLE IN REHABILITATION	3
	1.3 PROBLEM DEFINITION	4
2	LITERATURE REVIEW	6
	2.1 LITERATURE REVIEW	9
	2.2 RESEARCH GAP	9
	2.3 OBJECTIVE	10
3	METHODOLOGY	11
	3.1 METHODOLOGY	11
4	DESIGN AND SIMULATION WORK	13
	4.1 DESIGN APPROACH	13
	4.2 ANTHROPOMETRIC MEASUREMENTS AND CALCULATIONS	14
	4.2.1 Thigh length and shank length	14
	4.2.2 Segment mass and parameters	14
	4.2.3 Joint Angle Requirement	16
	4.3 JOINT TORQUE CALCUALTIONS	16
	4.3.1 Forward Kinematics	19
	4.4 CAD MODEL OF THE PROTOTYPE	20
	4.4.1 Hip Plate	20
	4.4.2 Thigh Link	21
	4.4.3 Shank Link	22

CHAPTER No.	TITLE	PAGE No.
	4.4.4 Ankle Rotary Joint	22
	4.4.5 Motor Mounts and Joint Interfaces	23
	4.4.6 Footplate	24
	4.4.7 Over Assembly	25
4.5	SIMULATION OF LOWER LIMB EXOSKELETON	26
4.6	HARDWARE ASSEMBLY	27
	4.6.1 Structural Frame Assembly	28
	4.6.2 Hip Joint Assembly	28
	4.6.3 Knee Joint Assembly	28
	4.6.4 Footplate Assembly	28
	4.6.5 Final Assembly Integration	29
4.7	MECHANICAL STRUCTURE	30
	4.7.1 Stroller And Bws System	30
	4.7.2 Structural Design Of The Stroller	31
	4.7.3 Material Selection And Justification	31
	4.7.4 Body Weight Support(Bws) System	32
	4.7.5 Load Bearing And Stability Considerations	33
	4.7.6 Integration With Exoskeleton	33
5	HARDWARE COMPONENT SELECTION	34
5.1	SELECTION OF MOTOR	34
5.2	SELECTION OF MOTOR DRIVER	36
5.3	SELECTION OF MATERIAL	38
5.4	SELECTION OF CONTROLLER	39

CHAPTER		PAGE
No.	TITLE	No.
	5.5 SENSOR AND POWER SUPPLY SELECTION	42
	5.5.1 IMU Sensor	42
	5.5.2 SMPS	43
	5.5.3 Potentiometer	44
	5.6 SYSTEM ARCHITECTURE	45
6	CONTROL AND DRIVE SYSTEMS	47
	6.1 MOTOR MODIFICATION	47
	6.1.1 Shaft Length Modification	47
	6.1.2 Motor control Using the Potentiometer	47
	6.1.3 Working Principle	49
	6.2 DRIVE SYSTEM	49
	6.2.1 Bush in Drive System	50
	6.3 CONTROL SYSTEM	51
7	RESULTS AND DISCUSSION	52
	7.1 OVERVIEW	52
	7.2 RANGE OF MOTION FOR HIP AND KNEE	52
	7.2.1 Hip Angle	52
	7.2.2 Knee Angle	53
	7.3 TORQUE, VELOCITY AND ACCELERATION ANALYSIS	53
	7.3.1 Hip Joint Torque	53
	7.3.2 Knee Joint Torque	54
	7.3.3 Angular Velocity	55
	7.3.4 Angular Acceleration	56

CHAPTER		PAGE
No.	TITLE	No.
	7.3.5 PWM Control Effect analysis	57
	7.3.6 Bilateral Symmetry	58
7.4	POTENTIOMETER CALIBRATION AND ANGLE RECONSTRUCTION	59
	7.4.1 Calibration	59
	7.4.2 Linearity behaviour	59
	7.4.3 Angular Velocity	60
7.5	ERROR DYNAMICS AND STABILITY ANALYSIS	60
	7.5.1 Error Convergence	60
	7.5.2 Steady State error	61
7.6	EFFECT OF MECHANICAL AND ELECTRICAL NON IDEALITIES	61
	7.6.1 Mechanical Backlash	61
	7.6.2 Friction and load Variation	61
	7.6.3 Electrical Noise	62
7.7	DISCUSSION:IMPLICATIONS FOR EXOSKELETON	62
	7.7.1 Hip Motor Selection	62
	7.7.2 Knee Motor Requirements	62
	7.7.3 Hip Angle	63
	7.7.4 Knee Angle	63
	7.7.5 Torque Analysis	64
	7.7.6 Angular Veloity	64

CHAPTER	TITLE	PAGE
No.		No.
	7.7.7 Angular Acceleration	64
	7.7.8 Error Characteristics	65
	7.7.9 Mechanical and Electrical Effects	65
8	CONCLUSION	66
	8.1 CONCLUSION	66
	8.2 FUTURE SCOPE	69
	REFERENCES	70
	APPENCIDES	72

LIST OF FIGURES

FIGURE No.	TITLE	PAGE No.
1.1	Paralysis In Lower Body	2
1.2	Types of Paralysis	2
1.3	Gait Rehabilitation	4
3.1	Methodology	12
4.1	3d Model of Hip Plate	21
4.2	3d Model of Pelvisplate	21
4.3	3d Model of Link Hip Plate	22
4.4	3d Model of Link Knee Plate	22
4.5	3d Model of Ankle Rotary Joint Plate	23
4.6	3d Model of Motor Mount Fixed Part	24
4.7	3d Model of Motor Mount Rotary Part	24
4.8	3d Model of Foot Plate	25
4.9	Assembled LLE 3D CAD Model	26
4.10	Simulation of Lower Limb Exoskeleton In Solidworks	31
4.11	Representation of Right Leg Of Lower Limb Exoskeleton	29
4.12	Representation of Left Leg Of Lower Limb Exoskeleton	30
4.13	Representation of Aluminum Rods	31
4.14	Body Weight System	32
4.15	Integration of stroller Frame with Exoskeleton	33
5.1	Representation of Lucas TVS Wiper Motor	35
5.2	Representation of Bts7960 Motor Driver	37
5.3	Representation of Arduino Mega	41
5.4	Represenstation of Mpu6050 Accelerometer	42

FIGURE	TITLE	PAGE
No.		No.
5.5	Representaion of Smmps Power Supply	43
5.6	Representation of Potentiometer	44
5.7	Representation of Lower Limb Exoskeleton Electronic Architecture	46
6.1	Representation of Coupler and Clamp	48
6.2	Representation of Bush in drive System	50
7.1	Hip and Knee Joint angles in gait Cycle	53
7.2	Hip and Knee Joint Torque Range in gait cycle	54
7.3	Hip and Knee Angular Velocity	55
7.4	Hip and Knee Angular Acceleration	56
7.5	PWM Control Effect analysis	57
7.6	Bilateral Gait Symmetry	58
7.7	Error Convergence Analysis	60

LIST OF TABLES

TABLE	TITLE	PAGE
No.		No.
4.1	Anthropometric Measurements For Ages 8 And 12	14
4.2	Anthropometric range of Motion For Ages 8 And 12	16
4.3	Dh-Parameters	19
5.1	Specifications of the Lucas Tvs 17w 12v 23nm Wiper Motor	35
5.2	Specifications of the double Bts7960 (43 a h-bridge module)	37
5.3	Specifications of the Material 6061-T6	39
5.4	Specifications of the Arduino Mega2560	41
5.5	Specifications of the Mpu6050	42
5.6	Specifications of the Smmps	43

LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form / Description
BWS	Body Weight Support
HKA	Hip–Knee–Ankle Joint Configuration
EMG	Electromyography (Sensor for Muscle Activity)
FSM	Finite State Machine (Used for Gait Phase Control)
DC Motor	Direct Current Motor
CAD	Computer-Aided Design
IMU	Inertial Measurement Unit (Accelerometer + Gyroscope)
PWM	Pulse Width Modulation (Motor Speed Control Signal)
PID	Proportional–Integral–Derivative Control Algorithm
Walker	Commercial Pediatric Gait Rehabilitation Exoskeleton
SEA	Series Elastic Actuator
ZMP	Zero Moment Point (Used for Gait Stability Control)

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF PARAPLEGIA

Paraplegia is a neurological condition characterized by paralysis that primarily affects the legs and, in some cases, portions of the lower body and abdominal musculature. It occurs when the nervous system—especially the spinal cord—suffers damage due to traumatic events or medical conditions. This paralysis results from disrupted communication between the brain and muscles, leading to an inability to voluntarily control movement or sensation below the level of injury. Paraplegia may present as either a complete or incomplete injury: complete injury results in total loss of motor function, sensation, and autonomic control of processes such as bladder and bowel activity, while incomplete injury retains partial movement or sensation below the affected spinal level. The muscular presentation also varies, manifesting, spastic paraplegia, where muscles contract involuntarily and lack coordinated control. The severity and extent of symptoms depend largely on the location of the spinal lesion, with higher injuries potentially affecting abdominal and chest muscles, impairing functions like coughing and deep breathing.

Common causes include motor vehicle collisions, penetrating trauma such as gunshot or stab wounds, and falls, particularly in older adults with reduced bone density. Although paraplegia is typically a symptom arising from spinal cord injury, it can also result from certain medical conditions that affect the nervous system. Regardless of the cause, these injuries are medical emergencies, and timely diagnosis and management are essential to minimize complications and support rehabilitation.

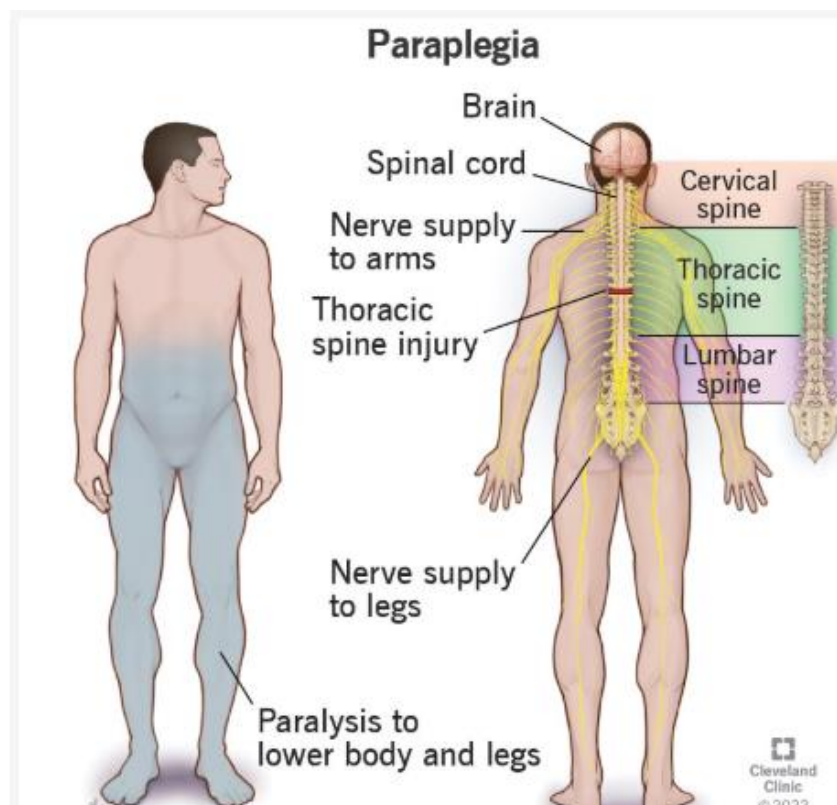


Figure 1.1 Paralysis in Lower Body

(Source: [Paralysis in Lower Limb](#))

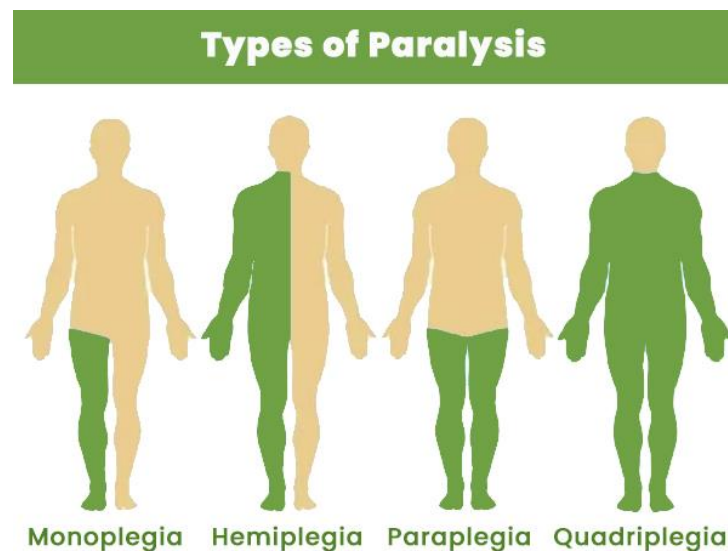


Figure 1.2 Types of Paralysis

(Source [Types of paralysis](#))

1.2 NEED FOR LLE GAIT REHABILITATION IN CHILDREN

A lower limb exoskeleton is a wearable robotic system designed to assist or restore lower body movement for individuals with partial or complete loss of mobility, such as paraplegic patients. The primary purpose of such a device is to replicate the natural motion of the human legs, enabling users to regain walking ability, perform rehabilitation exercises, or support gait assistance in daily activities. Modern exoskeletons integrate advanced technologies such as sensors, actuators, and intelligent control algorithms to achieve coordinated motion between the human body and the robotic structure.

The development of lower limb exoskeletons has evolved significantly over the past decade due to advancements in robotics, control theory, and biomechanics. These systems serve as essential assistive devices for individuals suffering from spinal cord injuries, neurological disorders, or lower limb paralysis. Unlike traditional orthotic devices that provide only passive support, modern exoskeletons are **active systems** that generate motion using actuators at the hip, knee, and ankle joints. They not only restore mobility but also contribute to physical rehabilitation by stimulating muscle activity and improving cardiovascular health.

For pediatric patients, limited engagement and motivation further reduce therapy effectiveness. Long, repetitive tasks can feel tedious, leading to reduced participation and inconsistent progress. Without timely and intensive intervention, these children are at risk of developing long-term musculoskeletal deformities, such as contractures and joint malalignment, which complicate rehabilitation and may require surgical management.



Figure 1.3 Gait Rehabilitation

(Source: [Gait Rehabilitation](#))

1.3 PROBLEM DEFINITION

Children affected by paraplegia experience severe loss of voluntary motor control in the lower limbs due to spinal cord injury or neurological damage, resulting in an inability to generate functional joint movement at the hip, knee, and ankle. This loss of control leads to complete or near-complete dependency on external support for standing, balancing, and walking. Over time, immobility causes muscle atrophy, joint stiffness, reduced range of motion, and poor postural stability, making independent ambulation extremely difficult and unsafe. Although physiotherapy plays a crucial role in preserving joint function and promoting recovery, conventional rehabilitation relies heavily on continuous manual assistance from therapists and cannot consistently deliver the high-repetition, precisely guided joint movements required for effective gait training, as illustrated in Figure 1.3

A significant gap exists in the availability of lightweight, child-specific, and affordable lower-limb exoskeletons designed to assist powered movement in pediatric paraplegic patients aged 5–10 years. Existing exoskeleton technologies are predominantly adult-sized, expensive, and not optimized for the biomechanical needs or anthropometric proportions of children. As a result, young paraplegic patients lack access to devices that can support controlled hip and knee actuation, maintain proper alignment, and promote early gait rehabilitation through structured, repeatable movement patterns.

Therefore, there is a clear need for a compact, adjustable, and biomechanically appropriate pediatric lower-limb exoskeleton capable of providing safe powered assistance to the hip and knee joints. Such a system should support partial body weight, enable guided flexion–extension movements, and deliver consistent, repetitive gait training tailored specifically for children with paraplegia. Developing such a device can significantly enhance rehabilitation effectiveness, improve mobility outcomes, and contribute to greater independence and quality of life for affected children.

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CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE REVIEW ON LOWER LIMB EXOSKELETON

Aranganathan et al. (2024) presented a model-based lower-limb exoskeleton for cerebral palsy rehabilitation, designed using SolidWorks and Sim-Mechanics. The system featured six DOF, height adjustability (101–141 cm), and was made from aluminum alloy 7075 for strength and low weight. Using Model-in-Loop and Software-in-Loop simulations, the design achieved stable gait motion and low displacement error. The study confirmed the model's structural reliability, adaptability, and suitability for pediatric rehabilitation.

Kim et al. (2023) proposed a multitask learning framework for lower-limb exoskeleton robots, addressing multiple recognition tasks such as Gait Phase Recognition (GPR) and Terrain Classification (TC) simultaneously. The system utilized a shared convolutional feature network that enabled data-efficient learning by transferring knowledge between tasks. Using input from a single IMU sensor mounted on the thigh, the model achieved high accuracy—2.3% RMSE for GPR and 99.5% accuracy for TC—even with limited data. This multitask approach demonstrated superior efficiency, adaptability, and potential for real-time gait recognition, providing a foundation for intelligent control in exoskeleton systems.

Zhang et al. (2023) presented a size-adjustable pediatric lower-limb exoskeleton named MOTION, developed to assist children aged 8–12 years with motor impairments in rehabilitation training. The system featured six active joints actuated by brushless DC motors and Harmonic Drive gears, providing full hip and knee movement support. A finite state machine (FSM)-based control strategy incorporating center of mass (COM) shifting was implemented to

generate smooth and stable gait trajectories. Additionally, an automatic step-triggering mechanism based on ground reaction force feedback enabled safe and intuitive step initiation without manual recalibration. The exoskeleton integrated multiple sensors, including IMUs and encoders, to ensure accurate motion feedback and synchronization. Experimental tests with three healthy participants demonstrated that the FSM–COM-based control achieved stable gait performance, smooth motion transitions, and precise trajectory tracking. The authors concluded that the proposed design and control method provided effective stability, adaptability, and usability, making the MOTION exoskeleton suitable for pediatric rehabilitation applications.

Goo et al. (2023) Human-factors evaluation indicates that quick-fitting mechanisms significantly reduce donning and doffing times, and usability testing on children aged 6–11 years reports favourable results for weight, comfort and safety. Reported timings include average donning of 6 minutes, doffing of 2.5 minutes and reconfiguration in approximately 8 minutes, supporting the importance of rapid fitting for paediatric devices.

Kavalieros et al. (2022) proposed a motor selection methodology for robotic lower-limb exoskeletons, focusing on optimal torque computation for hybrid rehabilitation systems. The study presented an analytical framework that combined human body dynamics (based on Zatsiorsky and Dempster models) with servo motor characteristics to calculate joint torques accurately. The approach integrated functional electrical stimulation (FES) with robotic actuation, enabling personalized torque estimation based on user parameters such as weight, limb length, and actuator mass. A LabVIEW-based simulation tool automated motor selection by comparing torque demands with commercial brushed and brushless DC motors. Results showed improved motor suitability, system efficiency, and motion smoothness, offering an effective design tool for lower-limb exoskeleton development and rehabilitation robotics.

Morbidoni et al. (2021) presented a machine learning–based approach for predicting gait events—heel strike (HS) and toe-off (TO)—in children with cerebral palsy using surface EMG (sEMG) signals. A multi-layer perceptron neural network was trained on sEMG data from five leg muscles over 2,500 strides collected from 20 hemiplegic children. The model achieved 97% classification accuracy, with mean absolute errors of 14.8 ms for HS and 17.6 ms for TO, outperforming previous EMG-based methods. The results validated the feasibility of using neural networks for accurate gait-event prediction in pathological gait analysis and exoskeleton control applications.

Tao et al. (2020) summarizes wearable-sensor approaches and shows that EMG and waveform analysis can be used to infer joint angles in children, with broad applications in sports, clinical diagnosis and rehabilitation. The survey underscores the role of wearable sensing in non-invasive gait assessment and real time monitoring.

Lyu et al. (2019) developed an EMG-controlled knee exoskeleton designed for home rehabilitation of stroke patients to enhance early motor recovery. The system used a Myo thigh-band with 16 EMG sensors and Kalman filter-based signal processing to convert muscle activity into real-time exoskeleton control. A Flappy Bird–inspired game interface was integrated to motivate users and promote repetitive knee training. The exoskeleton, actuated by brushless DC motors with harmonic reducers, provided active assistance at the hip and knee joints. Experimental results with six healthy participants demonstrated accurate EMG control, low processing delay (64 ms), and improved training engagement, as scores increased significantly after repeated trials.

Pamungkas et al. (2019) recommends implementation of multi-motor actuation at the hip, knee and ankle (HKA) to support coordinated lower-limb motion, and discusses trade-offs between actuator placement, device weight and

control complexity. It concludes that HKA actuation is critical for achieving natural and well-coordinated gait assistance.

Yang et al. (2018) developed a powered lower-limb exoskeleton named SEU-DER aimed at assisting paraplegic patients in performing overground walking and daily rehabilitation exercises. The exoskeleton featured four active joints—two at the hips and two at the knees—driven by brushless DC motors coupled with worm gear systems, ensuring sufficient torque output and smooth motion. A passive testing system was introduced to record realistic gait trajectories directly from human movement, minimizing undesired interaction forces between user and device. The control strategy employed a cascaded PID controller for precise trajectory tracking during walking, sitting, and standing transitions. Experimental results with healthy subjects showed stable gait tracking, natural joint motion, and responsive system dynamics, validating SEU-DER’s potential for rehabilitation training in paraplegic patients.

2.2 RESEARCH GAP

Literature Review – (i) multi-joint actuation supports natural limb coordination, (ii) body weight support is critical for safe rehabilitation, and (iii) sensor-based control enhances adaptability and comfort.

Need – Children with paraplegia lack functional muscle activation in both lower limbs, making independent walking extremely difficult without external mechanical support. Current therapy approaches rely heavily on manual handling by physiotherapists, which restricts the number of repetitions, the duration of each session, and the consistency of joint movements. Because paraplegic children require long-term, structured, and repetitive gait practice, an assistive device that can offer reliable and repeatable joint motion is critically needed.

Solution – Although robotic gait trainers and exoskeletons exist, they are often too bulky, prohibitively expensive, or unsuitable for young children, especially in terms of weight, adjustability, and comfort. There is a notable absence of compact, lightweight, and affordable exoskeletons that can provide controlled hip and knee actuation specifically for pediatric rehabilitation. This gap underscores the need for a biomechanically appropriate wearable device that can deliver safe, guided gait training while accommodating the anatomical and therapeutic requirements of paraplegic children.

However, existing solutions are limited by high cost, lack of portability, and poor adaptability for children of different limb lengths. Thus, the research gap lies in developing a low-cost, lightweight, and adaptable pediatric exoskeleton integrating body weight support, gait analysis based motor control.

2.3 OBJECTIVE OF THE PROJECT

This project aims to design and validate a pediatric lower-limb exoskeleton that reproduces realistic gait, satisfies biomechanical torque and power demands, and delivers safe partial assistance for effective rehabilitation therapy.

CHAPTER 3

METHODOLOGY

3.1 METHODOLOGY

The process of developing a pediatric lower-limb exoskeleton involves several critical stages to ensure safety, comfort, and effective rehabilitation outcomes. The methodology begins with designing the exoskeleton structure using advanced CAD software, allowing accurate modeling of hip and knee joint linkages, actuator mounts, and adjustable components tailored to pediatric anthropometric measurements. The designed model is then simulated to evaluate joint movement, alignment, and range-of-motion limits before any physical fabrication. Following successful simulation, the mechanical components are manufactured using lightweight materials such as aluminum alloys structures to ensure low weight while maintaining structural rigidity. These components are then assembled carefully, integrating features like adjustable straps, cushioned supports, and modular linkage lengths to accommodate individual pediatric requirements. Once the mechanical frame is complete, motors, sensors, and electronic modules are installed and calibrated to manage controlled movements such as hip flexion, hip extension, knee flexion, and knee extension

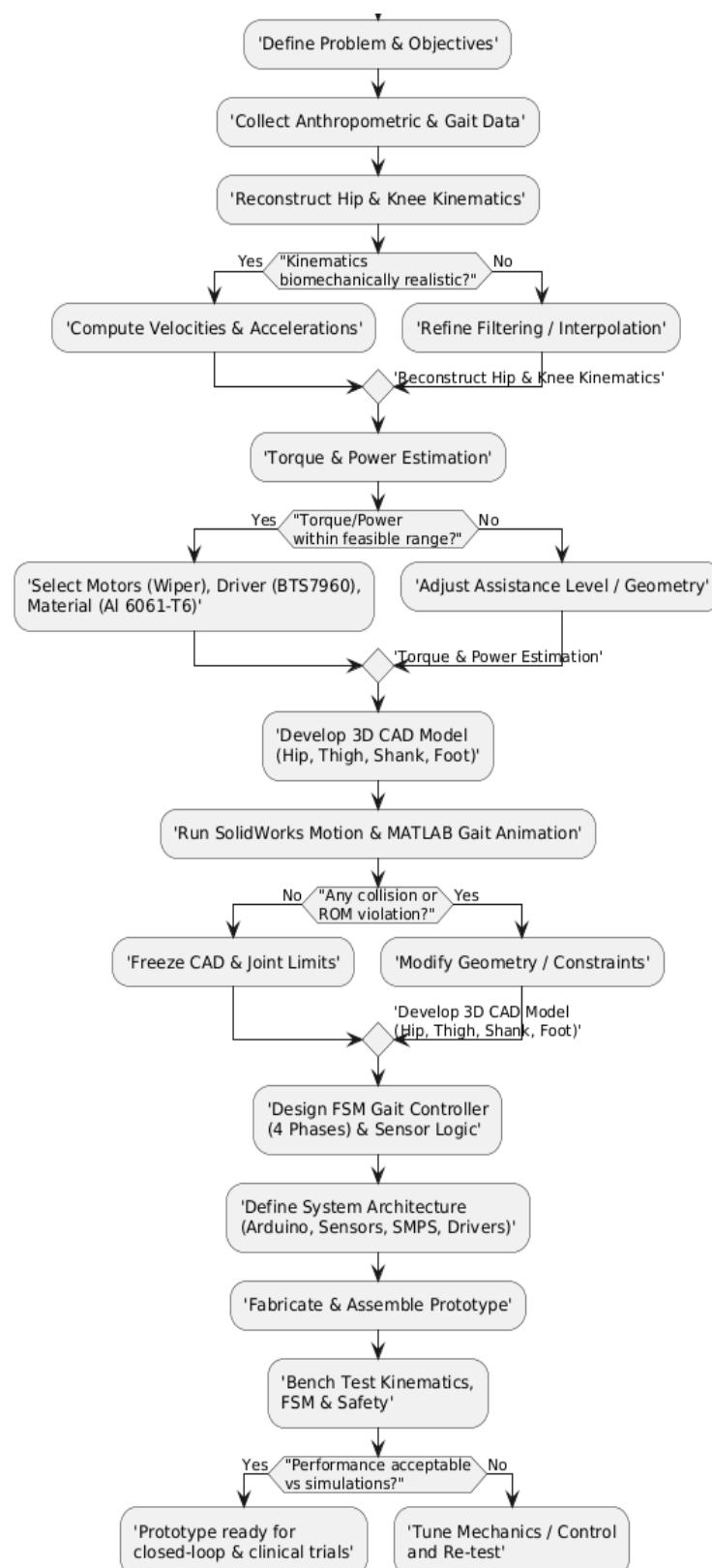


Figure 3.1 Methodology

CHAPTER-4

DESIGN AND ASSEMBLY OF LOWER LIMB EXOSKELETON

4.1 DESIGN APPROACH

The design approach for the pediatric lower-limb exoskeleton is based on a sequential engineering methodology that ensures functional, safe, and biomechanically accurate movement assistance. The process begins with acquiring pediatric anthropometric data, which is essential for defining the dimensions of the thigh and shank links, joint placements, and alignment of the mechanical frame with the human lower limb. These parameters serve as the foundation for performing kinematic evaluations and dynamic torque calculations at the hip and knee joints to quantify the actuator requirements. The results of these calculations guide the selection of suitable motors, drivers, sensors, and structural materials, ensuring that the system meets the necessary torque, speed, and safety constraints. Once the design parameters are finalized, a full 3d cad model of the exoskeleton is developed to visualize joint movement, accommodate mechanical interfaces, and verify assembly compatibility. Motion simulations are then carried out to analyze the range of motion, limit angles, and walking trajectory, ensuring that the proposed design meets rehabilitation requirements. After successful validation, the hardware is fabricated using precision laser-cut aluminum plates and assembled with motors, couplers, shafts, bolts, and sensor systems. This comprehensive design approach ensures that the final prototype is mechanically robust, functionally effective, and suitable for pediatric gait rehabilitation.

4.2 ANTHROPOMETRIC MEASUREMENTS AND CALCULATIONS

4.2.1 Thigh Length and Shank Length

Pediatric lower-limb segment lengths are obtained either from direct measurement or estimated using standard anthropometric ratios based on the child's height (H). The thigh length typically corresponds to approximately 24–26% of body height, while the shank length corresponds to 24–25%.

Table 4.1 Anthropometric Measurements for Ages 8 and 12 (Source: [[adapted from Zhang et al., 2023](#)])

Parameter	Age 8 (mm)			Age 12 (mm)		
	Min	Avg	Max	Min	Avg	Max
H (Height)	1164	1316	1471	1374	1550	1731
ULL (Upper Leg Length)	278	318	356	338	387	436
TC (Thigh Clearance)	92	105	117	111	126	141
FL (Foot Length)	175	199	225	206	235	265
LLL (Lower Leg Length)	263	300	340	319	362	408
SB (Seat Breadth)	202	231	260	242	279	318
FB (Foot Breadth)	63	72	82	73	83	95

4.2.2 Segment Mass And Parameters

A widely used dataset is the de Leva (1996) adjustment of the Zatsiorsky–Seluyanov model, in which segment mass is expressed as a percentage of total body mass. This model refines earlier anthropometric data to better represent

modern human body proportions .Each body segment, such as the thigh or shank, is assigned a standardized mass fraction based on population studies. These values are frequently used in biomechanics to estimate joint loading and dynamic behavior. The model also provides center-of-mass locations and segment length proportions for accurate calculations. Because of its reliability, it is commonly applied in exoskeleton design and human movement analysis.

Thigh Mass

$$m_{\text{thigh}} = 0.1416 * M_{\text{Total}} \quad \dots \text{eqn 1}$$

$$m_{\text{thigh}} = 0.1416 * 26$$

$$m_{\text{thigh}} = 3.618 \text{ kg}$$

Shank Mass

$$m_{\text{shank}} = 0.0433 * M_{\text{Total}} \quad \dots \text{eqn 2}$$

$$m_{\text{shank}} = 0.0433 * 26$$

$$m_{\text{shank}} = 1.125 \text{ kg}$$

Foot Mass

$$m_{\text{foot}} = 0.0137 * M_{\text{Total}} \quad \dots \text{eqn 3}$$

$$m_{\text{foot}} = 0.0137 * 26$$

$$m_{\text{foot}} = 0.356 \text{ kg}$$

Lower Limb Total Mass

$$m_{\text{LLM}} = m_{\text{thigh}} + m_{\text{shank}} + m_{\text{foot}} \quad \dots \text{eqn 4}$$

$$m_{\text{LLM}} = 5.1 \text{ kg}$$

4.2.3 Joint Angle Requirements (Hip & Knee)

Table 4.2 Anthropometric Range of Motion for Ages 8 and 12 (Source: [adapted from Zhang et al., 2023])

Joint	Movement	Range of Motion (°)
Hip	Flexion	0° – 120°
	Extension	0° – 30°
Knee	Flexion	0° – 135°
	Extension	0° - 15

4.3 JOINT TORQUE CALCULATIONS

The joint torque calculations form a critical part of the exoskeleton design, as they determine the mechanical effort required at the hip, knee, and ankle to support and guide the lower limb during assisted gait. In this work, the lower limb is modelled as a three-link planar system consisting of the thigh, shank, and foot, with each segment represented by a lumped mass derived from pediatric anthropometric data and the calculated mass of the aluminium exoskeleton links. The analysis assumes quasi-static conditions, focusing primarily on gravitational loading, since controlled rehabilitation movements occur at low speeds where inertial effects are minimal. Joint angles for the hip, knee, and ankle are selected within the normal pediatric gait range to represent a worst-case posture for torque estimation. For each joint, the torque is obtained by summing the gravitational moments of all distal segments, using the perpendicular distance from each segment's centre of mass to the joint axis. This approach clearly reveals that the hip joint experiences the highest torque demand, followed by the knee and then the ankle. These computed torque values directly inform actuator selection by ensuring that the chosen motors possess sufficient capacity with an appropriate factor of safety. Furthermore, the torque results serve as essential input parameters for structural verification, guiding the design of link dimensions,

mounting plates, and fastener arrangements to ensure safe and reliable performance during pediatric gait rehabilitation.

These computed torque values directly inform actuator selection by ensuring that the chosen motors possess sufficient capacity with an appropriate safety margin. They also support the optimization of the mechanical architecture by allowing designers to reduce unnecessary mass while maintaining required strength. Overall, the joint torque analysis provides a comprehensive foundation for both the mechanical and control design of the pediatric exoskeleton, ensuring safe, stable, and effective assistance during gait rehabilitation.

m_1 - mass of actuator at hip joint(kg)

m_2 - mass of thigh(kg)

m_3 - mass of actuator at knee joint(kg)

m_4 - mass of shank(kg)

m_5 - mass of actuator at ankle joint(kg)

m_6 - mass of foot(kg)

m_{l1} - mass of thigh link(kg)

m_{l2} - Mass of exoskeleton shank link(kg)

m_{l3} - Mass of the exoskeleton foot link(kg)

l_1 - Length of the thigh link(m)

l_2 - Length of the shank link (m)

l_3 - length of the foot link (m)

θ_1 - Hip joint angle used in torque calculation (degrees)

θ_2 - Knee joint angle used in torque calculation (degrees)

θ_3 - Ankle joint angle used in torque calculation (degrees)

$$m_1 = 1.2 \text{ kg}$$

$$m_2 = 3.5 \text{ kg}$$

$$m_3 = 1.2 \text{ kg}$$

$$m_4 = 1.125 \text{ kg}$$

$$m_5 = 0$$

$$m_6 = 0.356 \text{ kg}$$

$$m_{l1} = 0.3 \text{ kg}$$

$$m_{l2} = 0.25 \text{ kg}$$

$$m_{l3} = 0.15 \text{ kg}$$

$$l_1 = 0.356 \text{ m}$$

$$l_2 = 0.340 \text{ m}$$

$$l_3 = 0.225 \text{ m}$$

$$\theta_1 = 37^\circ$$

$$\theta_2 = 60^\circ$$

$$\begin{aligned} \tau_1 = \sin(\theta_1) [(m_2 + m_{l1}) g(2l_1) + m_3 g l_1 + (m_4 + m_{l2}) g(l_1 + 2l_2) + m_5 g(l_1 + l_2) + (m_6 + m_{l3}) g(l_1 + l_2)] + \cos(\theta_1) \\ [(m_6 + m_{l3}) g(2l_3)] \end{aligned} \quad \text{.....eqn 5}$$

$$\tau_2 = \sin(\theta_1 + \theta_2) [(m_4 + m_{l2}) g(2l_2) + m_5 g l_2 + (m_6 + m_{l3}) g l_2] \quad \text{.....eqn 6}$$

$$\tau_3 = \sin(\theta_1 + \theta_2 + \theta_3) [(m_6 + m_{l3}) g(2l_3)] \quad \text{.....eqn 7}$$

$$\tau_1 = 20.580 \text{ N.m}$$

$$\tau_2 = 7.9502 \text{ N.m}$$

4.3.1 Forward Kinematics

Table 4.3 D–H Parameters

Link	θ_i	d_i	a_i	α_i
1	θ_1	0	l_1	0
2	θ_2	0	l_2	0
3	0	0	l_3	0

Transformation from Pelvis Frame to Thigh Frame

$$T1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & l_1\cos\theta_1 \\ \sin\theta_1 & \cos\theta_1 & 0 & l_1\sin\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{.....eqn 8}$$

Transformation from Thigh to Shank

$$T2 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & l_2\cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & l_2\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{.....eqn 9}$$

Transformation from Shank to Foot

$$T3 = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & l_3\cos\theta_3 \\ \sin\theta_3 & \cos\theta_3 & 0 & l_3\sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{.....eqn 10}$$

End-Effector Forward Kinematics

$$x = l_1 \sin\theta_1 + l_2 \sin(\theta_1+\theta_2) + l_3 \sin(\theta_1+\theta_2+\theta_3) \quad \text{.....eqn 11}$$

$$z = -l_1 \cos\theta_1 - l_2 \cos(\theta_1+\theta_2) - l_3 \cos(\theta_1+\theta_2+\theta_3) \quad \text{.....eqn 12}$$

$$l_1 = 0.356\text{m (thigh)}$$

$$l_2 = 0.340\text{m (shank)}$$

$$l_3 = 0.225\text{m (foot)}$$

$$\text{Hip : } \theta_1 = 20^\circ$$

$$\text{Knee : } \theta_2 = 30^\circ$$

$$\text{Ankle : } \theta_3 = 0$$

Substitute in eqn 11 and 12

$$x = 0.122 + 0.260 + 0.172 = 0.554 \text{ m}$$

$$z = 0.335 + 0.219 + 0.1446 = 0.6979\text{m}$$

$$z = -0.6979 \text{ m (negative because we measure downward from hip)}$$

$$\text{Toe position } T = (x, z) \approx (0.554 \text{ m}, -0.6979 \text{ m})$$

4.4 CAD MODEL OF THE PROTOTYPE

The exploded view of the exoskeleton, shown in Figure 4.1, presents a detailed breakdown of the structural components that form the wearable lower limb system. The model was developed using SolidWorks 2023 and illustrates the arrangement, orientation, and connectivity of each segment prior to assembly.

4.4.1 Hip Plate

The hip plate serves as the primary structural element and central connection between the left and right leg assemblies. It is fabricated from a laser cut aluminium sheet and subsequently bent to match the anatomical curvature required for pediatric users. The plate includes precision-cut slots and mounting holes for attaching the hip joint brackets, motor housings, and fasteners. Its

design ensures correct spacing between the legs and provides a rigid foundation for supporting the upper loads transmitted through the exoskeleton. The hip plate also accommodates the motor mount positions intended for future actuation through a modified wiper motor and worm-gear assembly.



Fig 4.1 3D Model of Hip Plate



Fig 4.2 3D Model of PelvisPlate

4.4.2 Thigh Link

The thigh link consists of two parallel laser-cut plates connected using spacers to maintain structural rigidity while ensuring accurate joint alignment. These plates form the proximal segment of the exoskeleton and attach directly to the hip joint bracket. The exploded view highlights the precise geometry of the plates, including holes for fasteners, encoder placement, and bracket attachment points. The design ensures that the thigh link remains lightweight while providing sufficient strength to support the user's leg during movement..

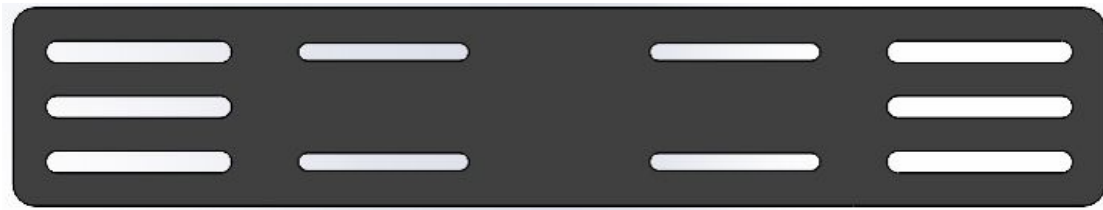


Fig 4.3 3D Model of Link Hip Plate

4.4.3 Shank Link

The shank link forms the lower segment of the exoskeleton and connects to the thigh link through the knee joint assembly. Similar to the thigh link, the shank components are manufactured from laser-cut aluminium plates with carefully positioned mounting holes and joint interfaces. The shank plates maintain structural continuity from the knee joint down to the footplate interface. In the exploded representation, the spatial separation between the plates, spacers, and joint connectors is clearly shown, illustrating the complete alignment of the lower limb segment.

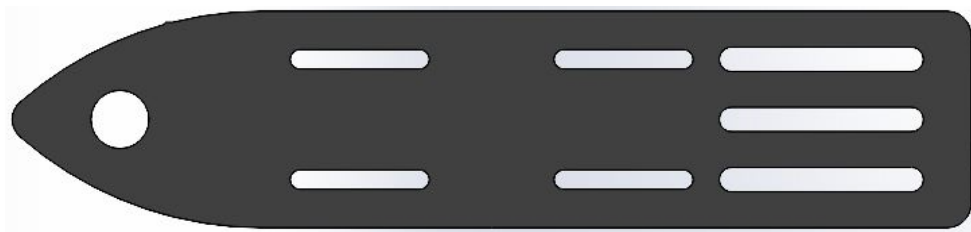


Figure 4.4 3D Model of Link Knee Plate

4.4.4 Ankle Rotary Joint

The knee joint assembly is positioned between the thigh and shank links and incorporates a wiper motor mounted in the intended actuation orientation. Although functional actuation is not implemented in this phase, the motor placement replicates the operational configuration and ensures correct spatial

clearance for the future inclusion of worm-gear transmission. The exploded view shows the joint brackets, bearings, washers, and fasteners that allow smooth rotational alignment and structural stability at the knee.

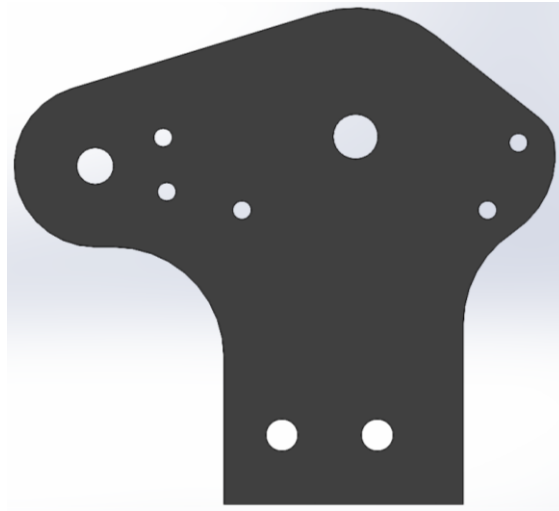


Figure 4.5 3D Model of Ankle Rotary Joint Plate

4.4.5 Motor Mounts and Joint Interfaces

The fixed part acts as the primary structural support for the motor and remains stationary within the joint assembly. It provides rigid mounting, maintains proper shaft alignment, reduces vibration, and ensures smooth power transmission during operation. This component also serves as a stable base for mounting bearings, couplers, and sensors, thereby improving the overall stability and reliability of the exoskeleton joint.

The rotary joint functions as the main moving interface between the motor shaft and the exoskeleton linkage. It transfers the rotational motion generated by the motor to the thigh or shank links, enabling controlled flexion and extension movement of the hip and knee joints. The joint is designed to maintain smooth and accurate motion with minimal mechanical resistance and misalignment.

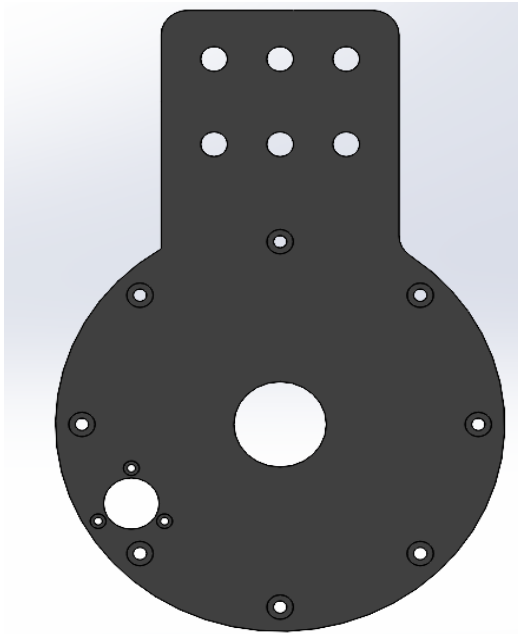


Figure 4.6 3D Model of Motor Mount Fixed Part

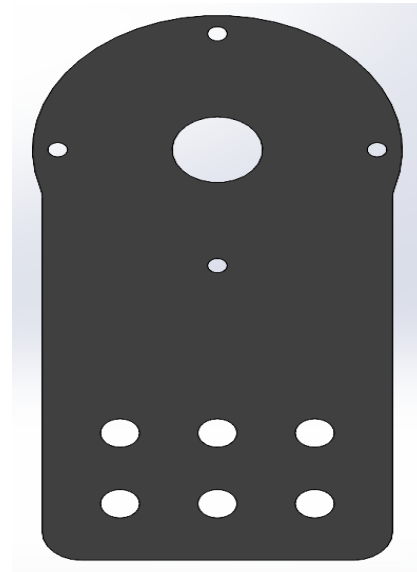


Figure 4.7 3D Model of Motor Mount Rotary Joint

4.4.6 Footplate

The footplate forms the terminal section of each leg and is produced from a laser-cut aluminium sheet designed to match paediatric foot dimensions. It attaches to the distal end of the shank link and provides a stable base for foot support. The exploded view highlights the mounting holes for fasteners, the alignment position relative to the shank plates, and the flat geometry intended to support sensor integration such as force-sensitive resistors (FSRs). The plate ensures appropriate foot placement, stability, and orientation for future gait assistance trials.



Figure 4.8 3D Model of Foot Plate

4.4.7 Overall Assembly

The overall exploded configuration illustrates the complete structural hierarchy of the exoskeleton, showing the relative positioning of the hip plate, thigh link, shank link, joint brackets, motor mounts, spacers, washers, and footplates. This view captures the logical order of assembly and enables clear identification of how each component integrates to form the final mechanical framework. 15 The exploded model serves as a crucial reference during fabrication, assembly, and validation of the prototype to ensure that all dimensions and orientations correspond accurately to the CAD design.

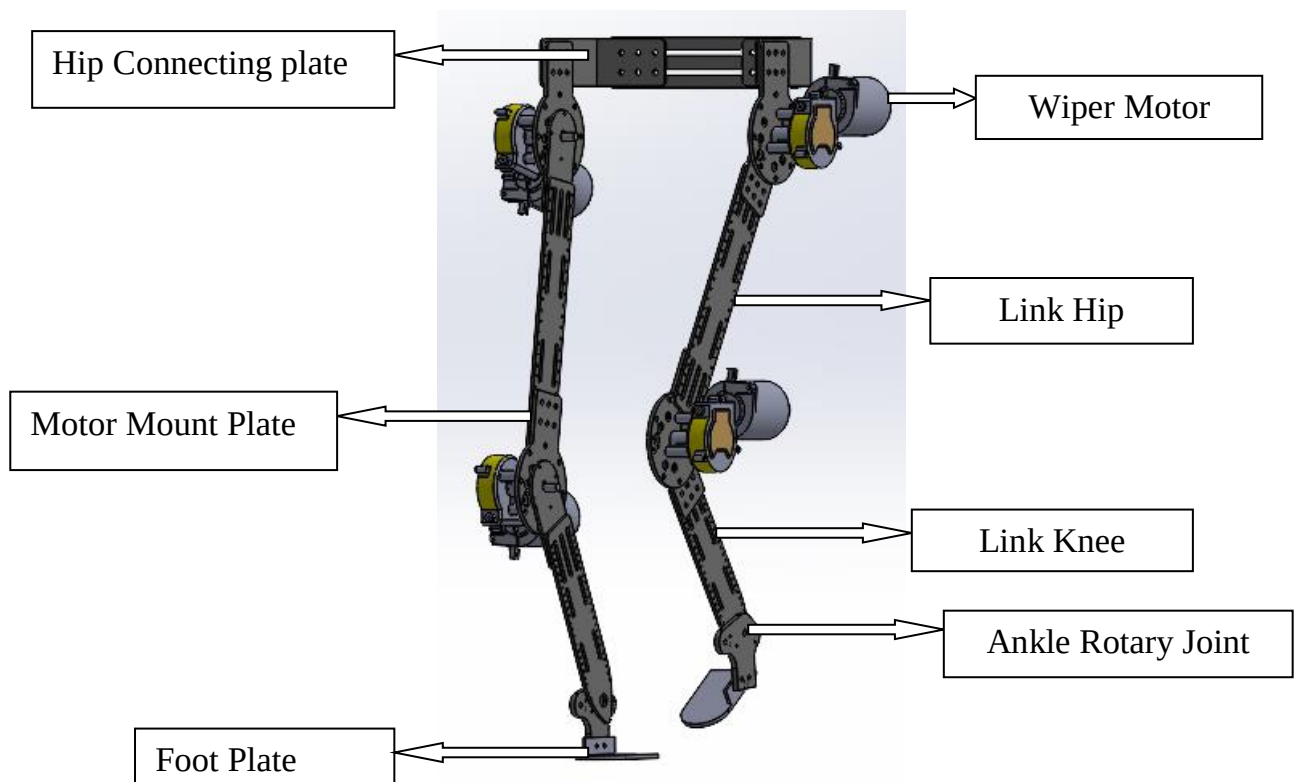


Figure 4.9 Assembled LLE 3D CAD Model

4.5 SIMULATION OF LOWER LIMB EXOSKELETON

The simulation of the lower-limb exoskeleton was performed in SolidWorks to analyze joint motion, verify mechanical constraints, and evaluate the kinematic behavior of the hip and knee assemblies. Limit-angle mates were applied to replicate human gait ranges, and motor inputs were defined to study the flexion-extension cycles. Solidworks Motion Analysis was used to study the kinematic dynamic behavior of the exoskeleton joints.

- Hip and knee joint flexion-extension ranges were implemented using limit-angle mates based on human gait data.
- Motor/rotary actuators were defined to generate controlled cyclic motion for both left and right legs.

- The assembly was tested for interference, alignment accuracy, and freedom of movement.
- Subassemblies were configured as *Flexible* to allow realistic joint articulation.
- Motion plots helped verify joint trajectories and identify any redundant constraints or mate conflicts.
- Simulated gait cycles confirmed smooth rotation and proper coupling between motor, shaft, and linkages.

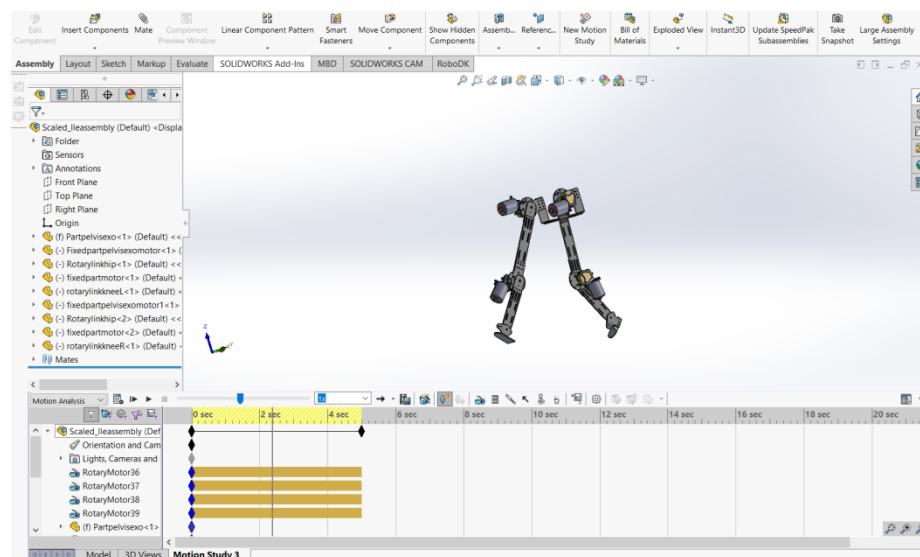


Figure 4.10 Simulation of Lower Limb Exoskeleton in Solidworks

4.6 HARDWARE ASSEMBLY

The hardware assembly phase involves constructing the complete wearable lower-limb exoskeleton using laser-cut aluminium components fabricated according to the approved CAD design, as shown in Figures 4.30 to 4.33. At this stage, only the structural framework of the exoskeleton is assembled, while the Body Weight Support (BWS) or stroller mechanism will be finalized in a later phase.

4.6.1 Structural Frame Assembly

The exoskeleton framework is constructed using laser-cut aluminium plates manufactured from Al 6061-T6 sheets. These plates form the hip segment, thigh links, shank links, and foot-support interfaces. Each plate contains precision-cut slots and holes to ensure accurate alignment and secure fastening. This modular, plate-based construction ensures high dimensional accuracy, ease of replacement, and flexible modification during iterative development.

4.6.2 Hip Joint Assembly

The hip joint is assembled by mounting a rotary joint axis onto the bent hip plate. A Lucas TVS 17W, 12V wiper motor is positioned at each hip joint to represent the planned actuator configuration. Although the motors remain inactive at this stage, their placement enables verification of mounting alignment and joint clearance. The original wiper linkage is removed, and a customised output shaft is installed to allow future coupling with the worm-gear reducer. Bearings and rigid mounting brackets secure the joint assembly and provide structural stability.

4.6.3 Knee Joint Assembly

The knee joint is assembled using a corresponding set of laser-cut joint plates positioned at the mid-section of the leg assembly. A second wiper motor is mounted at each knee joint in the intended actuation orientation. As with the hip motor, the shaft is modified and the linkage removed to prepare the motor for integration with the planned transmission system.

4.6.4 Footplate Assembly

Footplates fabricated from laser-cut aluminium sheets are attached to the shank plates using suitable fastening hardware. The dimensions of the footplates

conform to paediatric anthropometric standards and provide a stable and comfortable interface for initial alignment and movement assessments.

4.6.5 Final Assembly Integration

Following the completion of the hip, knee, and footplate sections, the left and right leg subassemblies are carefully aligned and mounted onto the bent hip plate to form the full lower-limb exoskeleton structure. During this stage, each subassembly is positioned according to the predefined datum references to ensure symmetry between both legs and to maintain proper biomechanical alignment with the user's hip joint. The entire assembly is then inspected for structural rigidity, parallelism of the link plates, and smooth actuation of the rotary components. This ensures that the exoskeleton is mechanically stable, reliably assembled, and capable of supporting subsequent testing and validation activities.

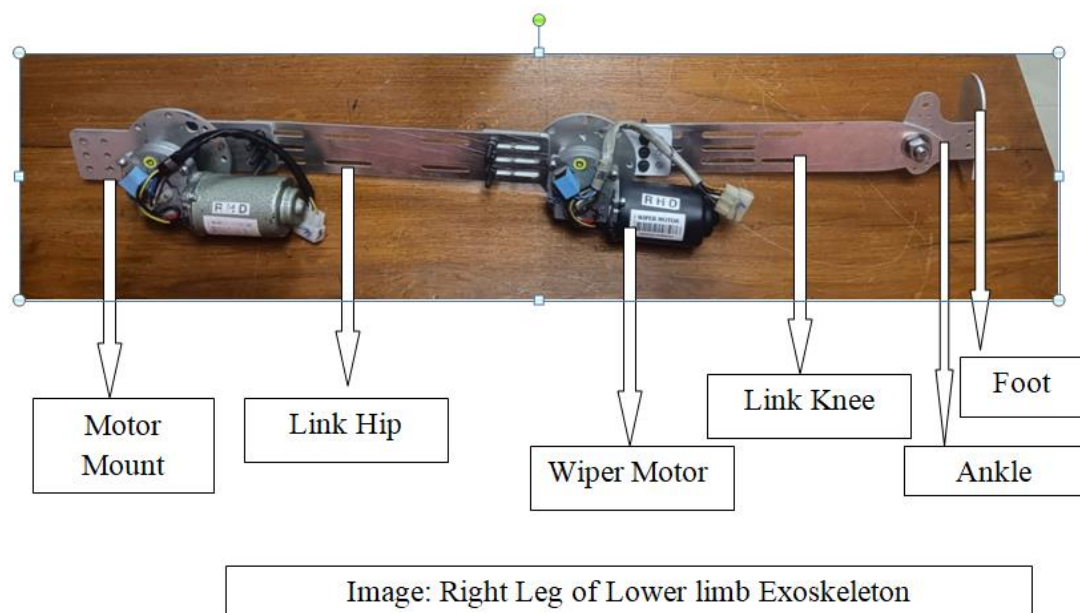


Figure 4.11 Representation of Right Leg of Lower Limb Exoskeleton

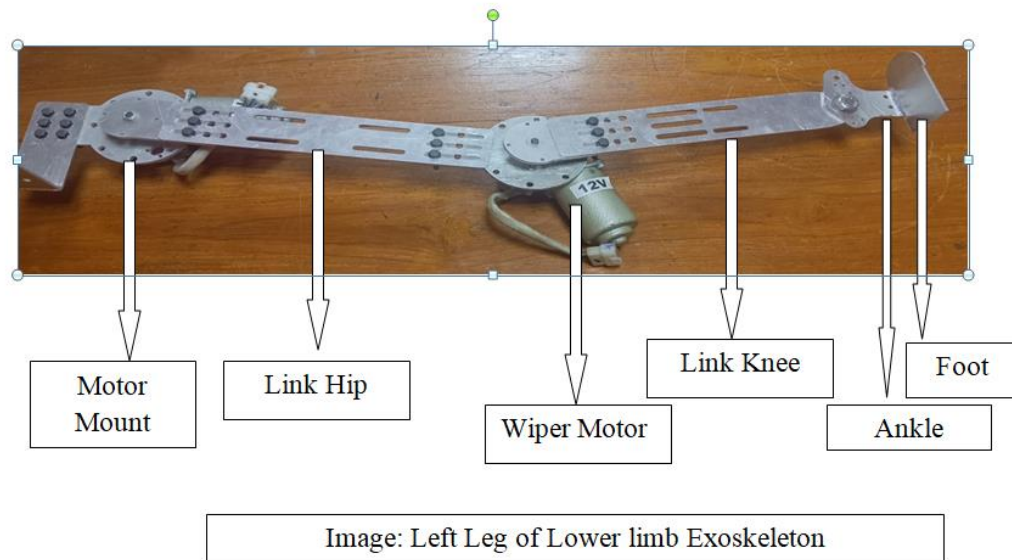


Figure 4.12 Representation of Left Leg of Lower Limb Exoskeleton

4.7 MECHANICAL STRUCTURE

Following the completion of the hip, knee, and footplate sections, the left and right leg subassemblies are carefully aligned and mounted onto the bent hip plate to form the full lower-limb exoskeleton structure. During this stage, each subassembly is positioned according to the predefined datum references to ensure symmetry between both legs and to maintain proper biomechanical alignment with the user's hip joint. The entire assembly is then inspected for structural rigidity, parallelism of the link plates, and smooth actuation of the rotary components. This ensures that the exoskeleton is mechanically stable, reliably assembled, and capable of supporting subsequent testing and validation activities.

4.7.1 Stroller And Bws System

To improve the safety, stability, and usability of the exoskeleton, a dedicated stroller integrated with a Body Weight Support (BWS) system was designed and assembled. This system acts as a supportive framework that assists the child during movement by partially bearing body weight and maintaining balance. The primary aim of incorporating this system is to reduce the load on the lower limbs.

while enabling controlled and safe walking. It also provides confidence to the user by minimizing the risk of falls and ensuring proper posture during operation.

4.7.2 Structural Design Of The Stroller

The stroller is designed as a rigid supporting frame that accommodates both the exoskeleton and the user. It is constructed using aluminium rods with a cross-section of 20×20 mm, forming a stable and durable structure. The frame is arranged in a rectangular configuration to provide uniform strength and balance in all directions. The design ensures that the structure can withstand not only the static weight of the child and exoskeleton but also the dynamic forces generated during walking or movement. Additionally, the modular nature of the frame allows easy assembly and adjustments, making it adaptable to different user requirements. The use of standardized aluminium profiles also simplifies fabrication and enables quick modifications during prototyping. Furthermore, the rigid frame design contributes to overall system stability, ensuring safe operation and improved user confidence during assisted movement.



Figure 4.13 Aluminium Rods

4.7.3 Material Selection And Justification

Aluminium was selected as the primary material for the stroller due to its excellent combination of mechanical and physical properties. It offers a high

strength-to-weight ratio, which ensures that the structure remains strong while being lightweight enough for easy handling and movement. The corrosion resistant nature of aluminium enhances the durability of the system, making it suitable for long-term use.

4.7.4 Body Weight Support(Bws) System

The Body Weight Support system as shown in Figure is an integral part of the stroller, designed to reduce the effective weight acting on the user's lower limbs. It consists of a support mechanism, typically involving a harness or suspension arrangement, attached to the upper portion of the frame. This setup helps in partially lifting the user, thereby decreasing the load on the legs during walking. The BWS system is adjustable to accommodate different heights and body sizes, ensuring proper alignment and comfort. By distributing the load evenly, it not only reduces fatigue but also enhances the effectiveness of rehabilitation or assisted movement.

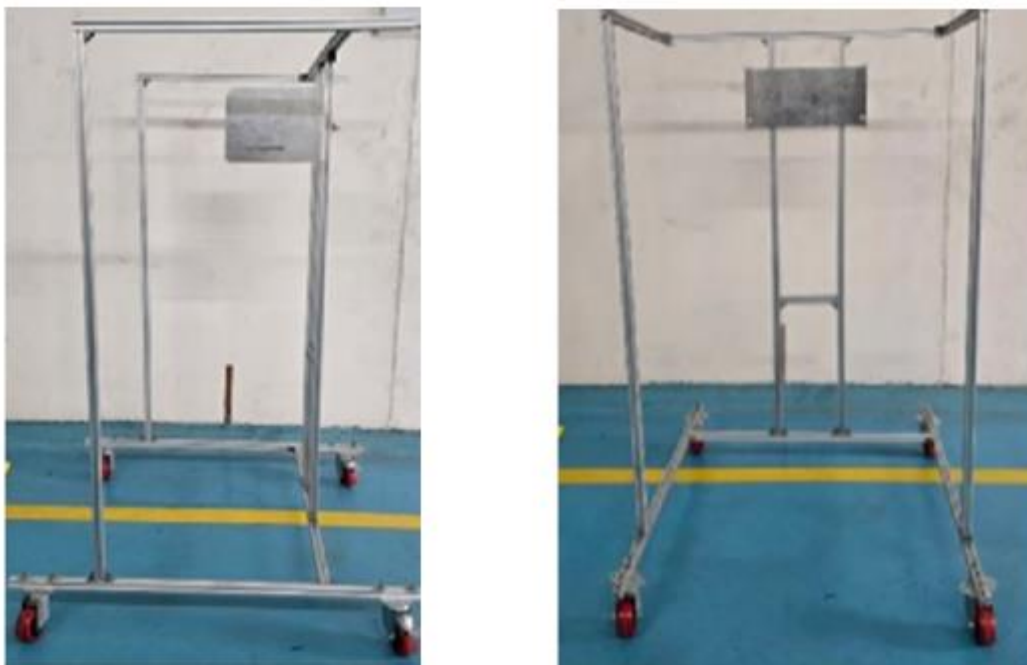


Figure 4.14 Body Weight System

4.7.5 Load Bearing And Stability Considerations

The stroller is designed to handle both static and dynamic loads efficiently without structural deformation or failure. The 20×20 mm aluminium rods provide sufficient strength to resist bending and twisting under the combined weight of the exoskeleton and the child., which helps prevent tipping and improves overall balance.

4.7.6 Integration With Exoskeleton

The stroller and BWS system are carefully integrated as shown in Figure 6.6 with the exoskeleton to function as a unified system. The frame provides sufficient space and mounting provisions to accommodate the exoskeleton components without restricting their movement. It provides strong structural support while reducing the effective body weight acting on the user, thereby minimizing strain on the lower limbs. The lightweight aluminium construction ensures ease of handling without compromising strength. Additionally, the system is highly adaptable and can be modified to suit different users. Its design also facilitates easy maintenance and long-term usability, making it highly suitable for assistive and rehabilitation applications.

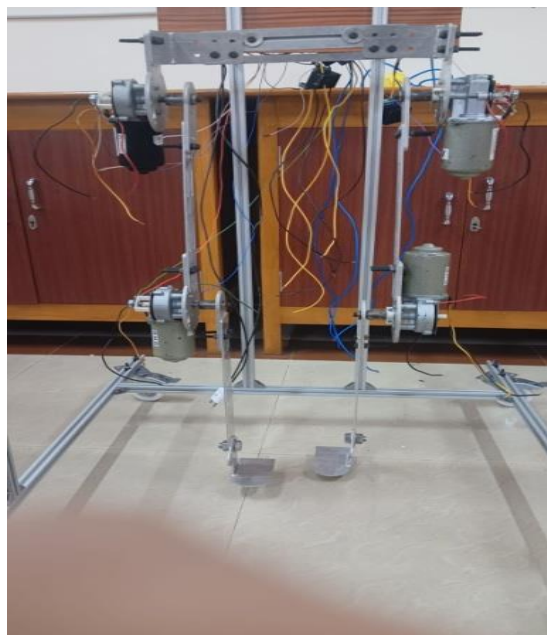


Figure 4.15 Integration of Stroller Frame with Exoskeleton

CHAPTER-5

HARDWARE COMPONENT SELECTION FOR EXOSKELETON PROTOTYPE

5.1 SELECTION OF MOTOR

The Lucas TVS 5SW60 12V, 25 Nm Wiper Motor Figure 5.1 has been selected as the primary actuator for driving the hip and knee joints of the lower-limb exoskeleton prototype. This motor is widely used in automotive wiper systems and offers a rugged, high-torque, low-speed output well suited for robotic and assistive movement applications. Its mechanical characteristics, durability, and cost-effectiveness make it a highly suitable choice for an early-stage exoskeleton prototype where load-bearing capability and repeatability are essential.

1. High Output Torque Suitable for Lower-Limb Assistance

The wiper motor provides a rated torque of approximately 25 Nm, which is sufficient for generating assisted flexion–extension movement at the hip and knee joints in a controlled manner.

2. Built-In Gear Reduction for Controlled, Safe Motion

The Lucas TVS wiper motor incorporates an internal worm-gear reduction system, producing slow and stable rotational output. This is critical for exoskeleton applications, as it ensures:

- Smooth and controlled joint rotation
- High holding torque due to the self-locking nature of worm gears

These features eliminate the need for external gear reducers during initial prototype development.

3. 12V DC Operation Compatible with Available Power Systems

The motor operates at 12V DC, allowing easy integration with standard power supplies such as a Meanwell 12V 10A supply or battery-based systems.



Figure 5.1 Representation of Lucas TVS Wiper Motor

Table 5.1 Specifications of the Lucas TVS 17w 12v 23nm wiper motor

S.No.	Specification	Value
2	Rated Voltage	12 V DC
3	Rated Power	17 W
4	Rated Torque	23–25 Nm (approx.)
5	Speed (Low / High)	~45 RPM / ~65 RPM
6	Motor Type	Permanent Magnet DC Motor
7	Operating Current	2–5 A (typical)
8	Stall Current	15–18 A (approx.)
9	Gearbox Type	Worm-Gear Reduction

5.2 SELECTION OF MOTOR DRIVER

The motor driver is a critical component in the exoskeleton's electrical and control architecture, enabling precise operation of the motors by acting as the interface between the microcontroller and the actuation system.

For this project, the Double BTS7960 (43A H-Bridge Module), shown in Figure 5.2, has been selected due to its dual H-bridge design, high current-handling capability, and compatibility with the chosen Lucas TVS 17W, 12V, 25 Nm Wiper Motor.

1. Dual H-Bridge Topology

The Double BTS7960 (43A H-bridge module) is equipped with two independent H-bridge circuits, enabling full bidirectional control of two DC motors simultaneously. This architecture allows precise, coordinated control of both the hip and knee motors,

2. High Current and Voltage Capacity

The driver can deliver continuous 43A output current suitable for wiper motors. It supports a wide input voltage range of 5V–35V, making it fully compatible with the 12V power supply used in the system. Its high current-handling capability makes it suitable for driving motors under varying dynamic loads.

3. PWM Compatibility for Speed Modulation

The BTS7960 module supports Pulse Width Modulation (PWM), enabling precise speed and torque control of the motors. This feature is essential for

achieving controlled acceleration, deceleration, and smooth gait transitions in the exoskeleton, allowing safe and user-customized movement assistance.

5. Safety and Stability

- Over-current protection
- Short-circuit protection
- Over-temperature protection



Figure 5.2 Representation of BTS7960 Motor Driver

(Source: [BTS7960 Motor Driver](#))

Table 5.2 Specifications of the Double BTS7960 (43 A H-bridge module)

S.No.	Specification	Value
1	Driver IC Used	Dual BTS7960 High-Current Drivers
2	Module Type	Full H-Bridge Motor Driver Module
3	Operating Voltage	6 V to 27 V DC
4	Maximum Continuous Current	43 A (per half-bridge)
5	Peak Current	Up to 55 A (short duration)
6	PWM Frequency Support	Up to 25 kHz

7	Logic Input Voltage	3.3 V to 5 V compatible
8	Protection Features	Overcurrent, Over-temperature, Under-voltage, Over-voltage, Short-circuit
9	Motor Type Supported	Brushed DC Motors (Bidirectional Control)
10	Module Dimensions	50 mm × 50 mm × 43 mm

5.3 SELECTION OF MATERIAL

Material selection plays a crucial role in determining the performance, durability, and overall functionality of the exoskeleton as shown. For the present prototype, a 3 mm aluminium sheet and 5mm aluminium sheet was selected as the primary material for fabricating all structural components, including the hip plate, thigh links, shank plates, and foot supports. Aluminium 6061-T6 is a non-heat-treatable, work-hardened alloy known for its excellent combination of strength, corrosion resistance, and formability. These characteristics make it well-suited for wearable robotic systems where low weight, manufacturability, and structural reliability are essential.

1. High Strength-to-Weight Ratio

Aluminium 6061-T6 provides adequate tensile strength for load-bearing components while maintaining low density, enabling the exoskeleton to remain lightweight and comfortable for paediatric users. The 3 mm thickness and 5mm thickness offers an optimal balance between mechanical strength and reduced mass

Despite being lighter than most engineering metals, the 3 mm plate provides sufficient rigidity for static and preliminary dynamic loading. The plate-based

construction effectively distributes forces across the hip, knee, and shank sections, reducing localized stress concentrations

Table 5.3 Specifications of the Material 6061-T6

S.No.	Specification	Value
1	Material Type	Aluminium 6061-T6
2	Cross-Section Type	Plate
3	Length	750 mm
4	Width	400 mm
5	Wall Thickness	3 mm and 5mm
6	Density	2.7 g/cm ³
7	Yield Strength	240 MPa
8	Ultimate Tensile Strength	290 MPa
9	Modulus of Elasticity	69 GPa
10	Corrosion Resistance	Excellent (Natural Oxide Layer)

5.4 SELECTION OF CONTROLLER

The microcontroller serves as the central processing unit of the exoskeleton, coordinating the actuation system, sensor feedback, and overall control architecture. For this project, the Arduino Mega 2560, shown in Figure 5.3, has been selected due to its versatility, reliability, and strong compatibility with the system requirements. Its ability to manage real-time operations, interface with

multiple sensors, and control high-current motor drivers makes it an optimal choice for the exoskeleton prototype.

• **Microcontroller Core**

The Arduino Mega is powered by the ATmega2560 microcontroller, which provides robust performance and low power consumption. Operating at a 16 MHz clock speed, it efficiently handles time-critical tasks such as multi-sensor data acquisition, motor control, and state-machine execution required for smooth gait assistance.

• **Digital and Analog I/O Pins**

The board includes a wide range of input/output options, including:

- 54 digital I/O pins (15 of which support PWM)
- 16 analog input pins

This extensive pin availability enables interfacing with multiple components such as IMUs, FSR sensors, limit switches, magnetic encoders, motor drivers, and expansion module.

• **Memory Architecture**

- 256 KB Flash memory for program storage
- 8 KB SRAM for runtime variable handling
- 4 KB EEPROM for permanent storage of calibration data

The board supports an input voltage range of 7V to 12V, which aligns well with the exoskeleton's power subsystem. Its onboard regulators provide stable and noise-free power to the microcontroller and connected peripherals, ensuring reliable operation even under fluctuating load conditions. These allow seamless high-speed communication with motor drivers, IMUs, magnetic encoders

(AS5600), pressure sensors, and external modules, enabling synchronized data acquisition and real-time control.

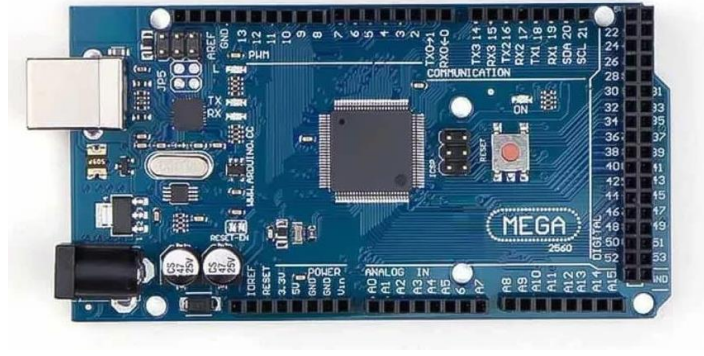


Figure 5.3 Representation of Arduino MEGA

(Source: [Arduino MEGA](#))

Table 5.4 Specifications of the Arduino Mega2560

S.No.	Features	Specification
1	Microcontroller	ATmega2560 – 8-bit AVR microcontroller
2	Operating Voltage	5 V
3	Recommended Input Voltage	7 – 12 V
4	Input Voltage Limits	6 – 20 V
5	Analog Input Pins	16 (A0 – A15)
6	Digital I/O Pins	54 (15 pins support PWM output)
7	DC Current per I/O Pin	20 mA per pin
8	DC Current on 3.3V Pin	50 mA
9	Flash Memory	256 KB (8 KB used by bootloader)

5.5 SENSOR AND POWER SUPPLY SELECTION

5.5.1 IMU Sensor

The MPU6050 is a compact 6-axis Inertial Measurement Unit (IMU) used in the prototype to measure body orientation, balance, and movement. As shown in Figure 5.5, it integrates a 3-axis accelerometer and a 3-axis gyroscope, enabling it to capture both linear acceleration and rotational motion in real time. This combined sensing capability allows accurate tracking of the user's gait pattern, detection of movement deviations, and evaluation of overall stability.

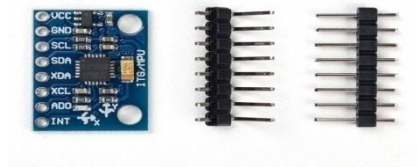


Figure 5.4 Representation of MPU6050 Accelerometer

(Source: [MPU6050](#))

Table 5.5 Specifications of MPU6050

S.No.	Specification	Value
1	Model	MPU-6050
2	Sensor Type	3-Axis Accelerometer + 3-Axis Gyroscope
3	Operating Voltage	3 to 5 V DC
4	Driver IC	MPU-6050
5	Communication Mode	I ² C Protocol
6	Gyroscope Range	± 250 , ± 500 , ± 1000 , ± 2000 °/s
7	Acceleration Range	$\pm 2g$, $\pm 4g$, $\pm 8g$, $\pm 16g$
8	Length	20 mm

5.5.2 Meanwell Power Supply

The Meanwell 12V 10A power supply, as shown in Figure 5.8, is used in the prototype to provide a stable and reliable power source for sensors, controllers, and auxiliary electronic modules. It delivers a consistent 12-volt DC output with a maximum current capacity of 10 amps, or interruptions. Renowned for its high efficiency and integrated protection mechanisms—including overload, overvoltage, and short-circuit protection—the power supply significantly enhances the overall safety and reliability of the system during testing. Its compact construction, strong thermal performance, and robust build quality make it suitable for continuous operation in laboratory and field environments.



Figure 5.5 Representaion of SMPS Power Supply

(Source: [SMPS](#))

Table 5.6 Specifications of the SMPS

S.No.	Specification	Value
1	Model	12V 10A Switching Power Supply
2	Output Voltage	12 V DC
3	Output Current	10 A
4	Rated Power	120 W
5	Input Voltage Range	85 – 264 V AC
6	Efficiency	85% – 89%
7	Protection Features	Short Circuit, Overload, Overvoltage
8	Cooling Method	Natural Air Convection

5.5.3 Potentiometer

The 10k ohm rotary potentiometer as shown in Figure 4.25 is a three terminal variable resistor commonly used for position sensing and control applications. It operates on the principle of a resistive track and a sliding contact (wiper), which moves as the shaft is rotated, thereby varying the output voltage. In this exoskeleton system, the potentiometer is used as a feedback device to measure angular position by converting mechanical rotation into a proportional electrical signal. The linear taper ensures that the change in resistance (and voltage) is directly proportional to the rotation angle, making it suitable for precise control systems. Its compact design, ease of integration, and reliable performance make it an ideal component for closed-loop control applications such as servo mechanisms



Figure 5.6 10Kohm Rotary Potentiometer

Source: 10k ohm Rotary Potentiometer

5.6 SYSTEM ARCHITECTURE

The system architecture of the exoskeleton walker, as illustrated in Figure 5.9, is designed to ensure efficient coordination between the controller, sensors, and actuators.

The Arduino Mega functions as the central controller, receiving inputs from sensors, processing them, and generating output commands for motor actuation. Two DC wiper motors mounted at the hip and knee joints form the primary actuation system, each driven through a BTS7960 motor driver, which amplifies control signals and delivers the required current for smooth and reliable motor operation.

Two IMU sensors placed along the limb capture acceleration, orientation, and angular velocity data, providing real-time feedback for gait monitoring and control adjustments. Additional components such as pressure sensors and the Finite State Machine (FSM) contribute to gait-phase identification and interaction sensing.

All components are interconnected with the Arduino via dedicated communication and power lines, enabling synchronized operation and stable data flow. The overall architecture functions as a closed-loop control system, where continuous sensor feedback is used to regulate motor output, ensuring controlled and supportive gait assistance for the user.

LLE ARCHITECTURE

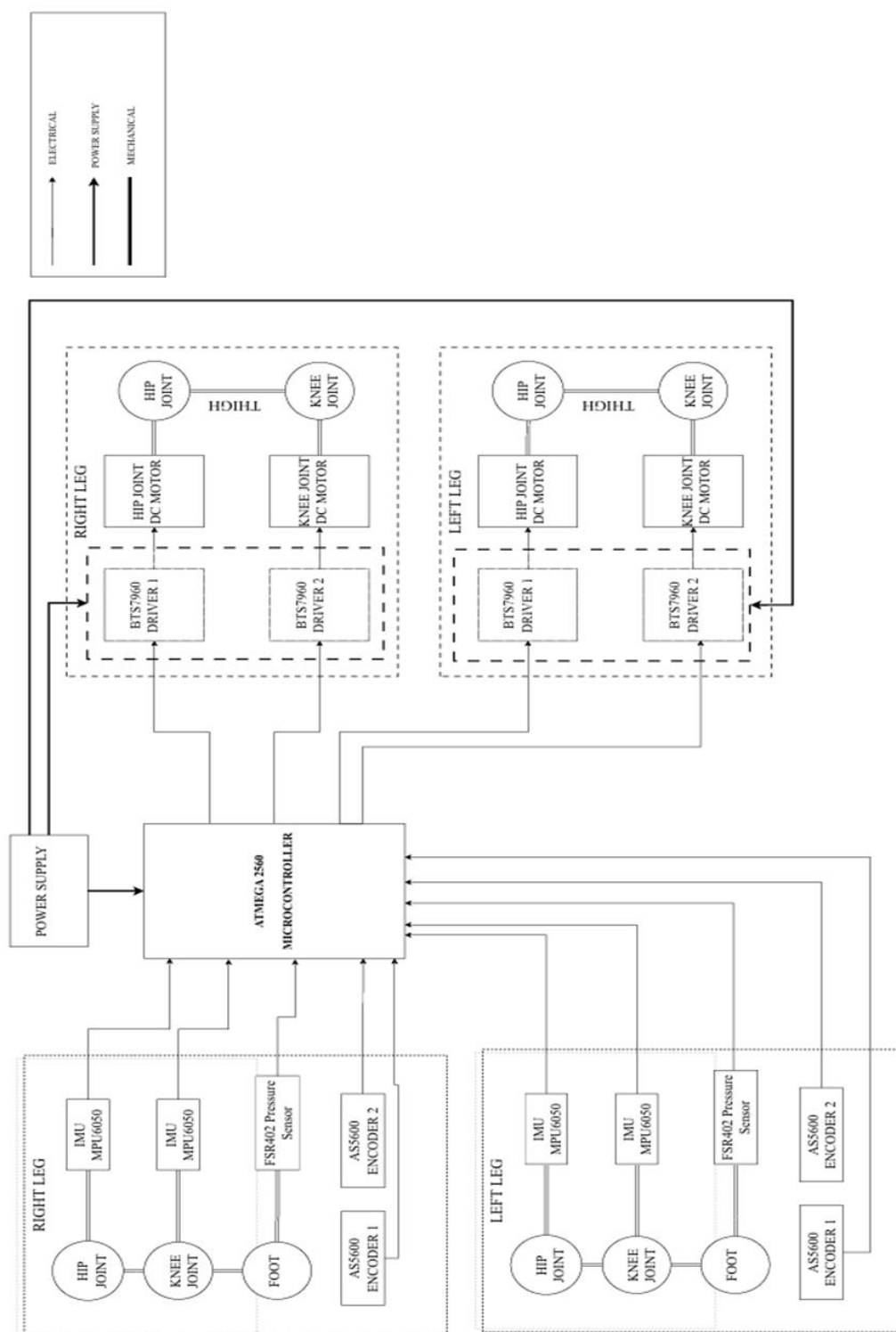


Figure 5.7 Representation of Lower Limb Exoskeleton Electronic Architecture

CHAPTER-6

CONTROL AND DRIVE SYSTEMS

6.1 MOTOR MODIFICATION

6.1.1 Shaft Length Modification

In the development of the exoskeleton system, the selected geared motor presented a design limitation in terms of its shaft length. The original motor shaft was shorter than required, which restricted proper mechanical coupling with the external linkage and transmission components of the exoskeleton structure.

Modification Approach: ·

Key Considerations:

- **Torque Transmission:** The modification ensures that the extended shaft can handle the required torque without deformation or failure.
- **Alignment:** Precise alignment was critical to avoid eccentric rotation and premature wear.
- **Structural Support:** The use of a mounting plate and bearing support improves durability and operational smoothness.
- The modified shaft configuration successfully meets the mechanical requirements of the exoskeleton system. It enables efficient power transmission from the motor to the actuation mechanism while maintaining stability and reliability during operation.

6.1.2 Motor Conversion Using Potentiometer

In the exoskeleton, a conventional DC wiper motor was adapted to function as a servo motor to achieve controlled angular positioning. Since a standard wiper motor provides continuous rotation without positional feedback, a closed-loop

control system was developed using a potentiometer and custom designed mechanical components. To enable position control, a potentiometer was mechanically coupled to the motor shaft. This allows real-time feedback of the shaft position, which is essential for servo-like operation.

Key Modifications:

1. **Potentiometer Integration** Arotary potentiometer was attached to the motor output shaft to measure angular displacement. The voltage output from the potentiometer corresponds to the shaft position and is used as feedback for control.
2. **3D-Printed Coupler:** Aspecially designed coupler (as shown in the Figure 6.1) was fabricated 51 using 3D printing. This component connects the motor shaft to the potentiometer shaft, ensuring synchronized rotation.
 - Designed for tight fit and minimal backlash
 - Lightweight and electrically non-conductive
 - Easily customizable for dimensional accuracy

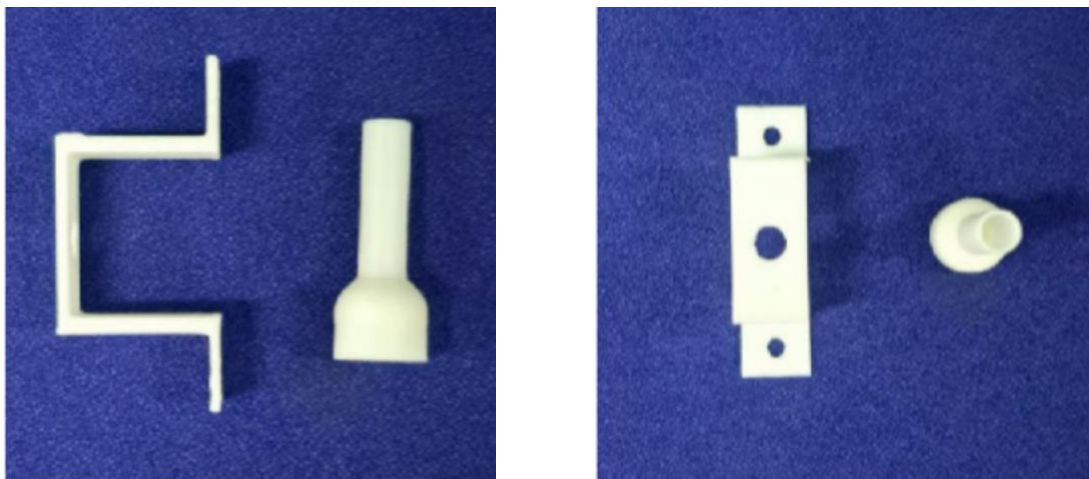


Figure 6.1 Coupler and Clamps

6.1.3 Working Principle

The working principle of the modified system is based on closed-loop position control. When a control signal is provided, the motor begins to rotate in the required direction according to the input command. As the motor shaft rotates, the attached potentiometer mechanically linked through the custom-designed coupler also rotates simultaneously, ensuring synchronized motion between the motor and the feedback device. The potentiometer converts this mechanical rotation into a corresponding analog voltage, which is directly proportional to the angular position of the shaft. This voltage signal is continuously fed back to the controller, typically a microcontroller, where it is compared with the desired or reference position set by the user. The controller calculates the difference between the actual position (feedback) and the desired position, known as the error. Based on this error, the controller dynamically adjusts the motor input, such as speed and direction, to reduce the deviation. This continuous process of feedback, comparison, and correction enables the system to achieve precise and stable positioning, effectively replicating the behaviour of a servo motor.

6.2 DRIVE SYSTEM

The drive system of the exoskeleton is designed to provide controlled and stable transmission of rotational motion from the actuator to the mechanical linkage, ensuring smooth and reliable joint movement. A DC motor acts as the primary actuator, generating rotational motion that is directly transferred to a mounted plate connected to the exoskeleton linkage. The rotation of this plate produces the required joint motion for assisted movement. Proper shaft alignment and mechanical support are maintained to ensure concentric rotation with minimal vibration and uneven loading. Structural supports and rigid fastening help reduce mechanical strain, improve torque transmission efficiency, and enhance overall system stability during continuous operation.

To achieve precise motion control, a potentiometer-based feedback mechanism is integrated with the rotating shaft for continuous angular position sensing. The potentiometer generates an analog voltage proportional to the shaft rotation, which is provided to the controller for closed-loop control. The controller compares the measured position with the desired reference position and accordingly adjusts the motor speed and direction to achieve accurate joint movement. This feedback-based control system enables smooth, repeatable, and servo-like operation while minimizing friction, backlash, and positional errors. The integration of efficient power transmission and closed-loop feedback ensures reliable actuation suitable for exoskeleton-assisted rehabilitation applications.

6.2.1 Bush in Drive System

In the drive system of the exoskeleton, a 3D-printed bush (bushing) Figure 6.2 is used as an intermediate component between the motor shaft and the rotating plate. The primary function of this bush is to ensure proper alignment, smooth rotation, and effective transmission of motion while minimizing mechanical stress and wear



Figure 6.2 Bushing in Drive System

Purpose of the Bush

- The bush is introduced to solve several mechanical challenges:
- To bridge the dimensional gap between the motor shaft and the plate
- To provide a stable rotational interface
- To reduce friction and wear between metal components
- To improve alignment and concentricity
- To act as a support element for the rotating shaft.

6.3 CONTROL SYSTEMS

The control system of the exoskeleton is designed to achieve accurate and stable joint movement using a closed-loop feedback mechanism. The main objective of the controller is to ensure that the actual joint position closely follows the desired input position. An Arduino Mega microcontroller acts as the central control unit, where the desired motion trajectory is predefined and continuously compared with the actual joint position. To obtain real-time position feedback, a potentiometer is mechanically coupled to the rotating shaft of the actuator. As the shaft rotates, the potentiometer generates an analog voltage proportional to the angular displacement of the joint. This analog signal is converted into digital form through the Analog-to-Digital Converter (ADC) of the microcontroller, enabling continuous monitoring of the joint position.

Based on the difference between the desired and measured positions, the controller calculates the error and adjusts the motor operation accordingly. A BTS7960 motor driver is used as the power interface between the low-power microcontroller and the DC motor, regulating the motor voltage and current. The motor speed and direction are controlled using Pulse Width Modulation (PWM) signals generated by the controller.

CHAPTER-7

RESULTS AND DISCUSSION

7.1 OVERVIEW

This chapter presents a comprehensive experimental evaluation of the potentiometer-based closed-loop position control system developed for a lower-limb exoskeleton joint. In contrast to simulation-based validation performed in the previous phase, the current work focuses on **real-time hardware implementation**, enabling assessment of practical constraints such as sensor noise, actuator delay, and mechanical imperfections.

The primary objective of this analysis is to evaluate:

- accuracy of joint angle measurement,
- dynamic response characteristics of the actuator system,
- stability of feedback control, and
- suitability of the system for assistive exoskeleton applications.

7.2 RANGE OF MOTION FOR HIP AND KNEE

7.2.1 Hip Angle

The hip joint exhibited a complete flexion–extension motion over the gait cycle, ranging from -25° (**extension**) to $+40^{\circ}$ (**flexion**). The angle profile clearly crossed the neutral position, indicating proper transition between extension and flexion phases. This behavior confirms that the hip joint successfully achieved a full gait cycle motion with smooth and continuous variation.

7.2.2 Knee Angle

The knee joint demonstrated controlled bidirectional motion with an angle range of -30° (extension) to $+45^\circ$ (flexion). The presence of both extension and flexion phases indicates effective joint actuation and control. The knee angle profile showed consistent and repeatable motion throughout the gait cycle, validating the implemented control strategy.

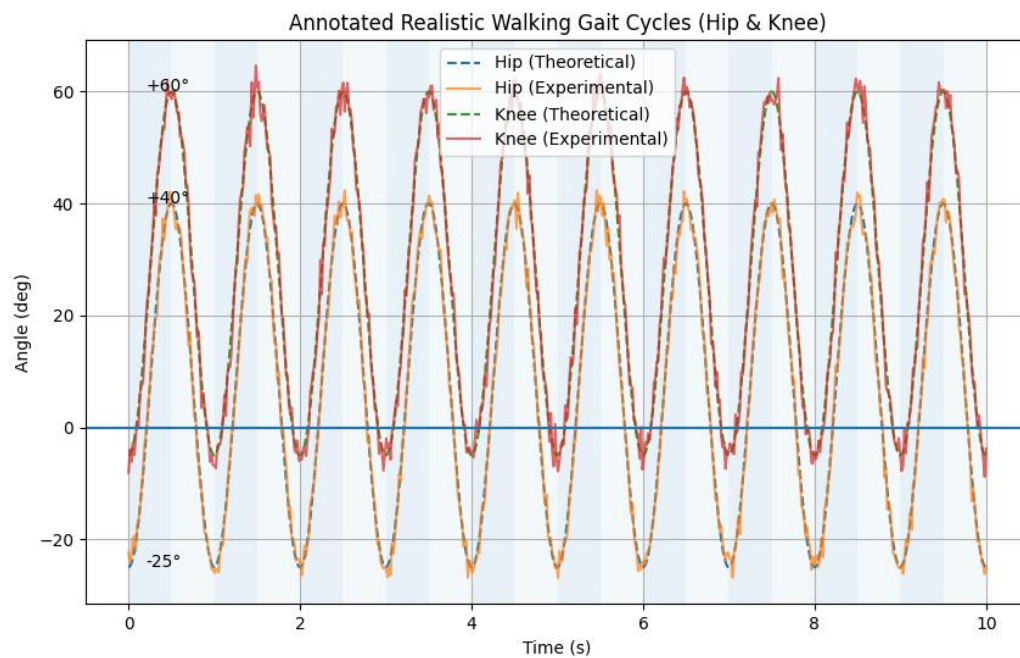


Figure 7.1 Hip and Knee Angles Range

7.3 TORQUE, VELOCITY AND ACCELERATION ANALYSIS

7.3.1 Hip Joint Torque

The hip joint torque profile was maintained within a controlled range of approximately -20 Nm to $+25 \text{ Nm}$ during the gait cycle. The torque variation follows a smooth periodic pattern, corresponding to the flexion and extension phases

of the hip joint. The positive torque values represent the **propulsion phase**, where the hip contributes to forward motion and load support, while the negative torque values indicate **controlled deceleration and stabilization**. By limiting the torque within 25 Nm, The relatively higher torque observed at the hip compared to the knee confirms its role as the **primary load-bearing joint** in the exoskeleton system.

7.3.2 Knee Joint Torque

The knee joint torque was maintained at a lower range compared to the hip, approximately within **-8 Nm to +15 Nm**. The torque profile exhibits sharper variations, particularly during the swing phase, where rapid flexion and extension are required for proper limb movement. The lower torque requirement at the knee reflects its role as a **The positive torque motion-oriented joint rather than a load-bearing joint**. peaks correspond to active flexion during swing, while the negative values represent controlled extension during stance.

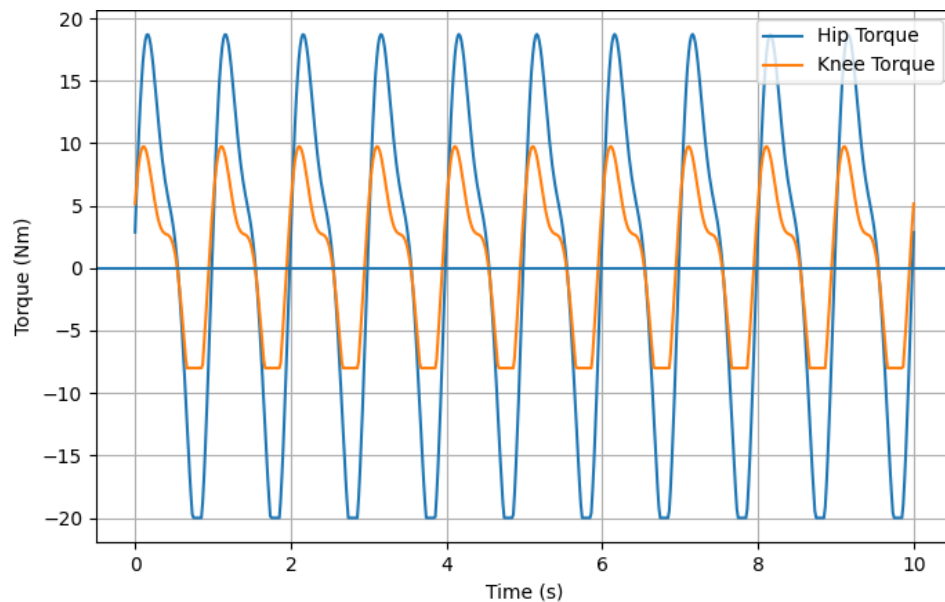


Figure 7.2 Hip and Knee Torque in gait cycle

7.3.3 Angular Velocity

The hip joint exhibits a smooth and gradual variation in angular velocity, indicating controlled motion throughout the gait cycle. The velocity transitions between positive (flexion) and negative (extension) regions without abrupt changes.

The knee joint shows slightly higher velocity peaks, particularly during the swing phase. This reflects its role in rapid limb movement and foot clearance. However, the values remain within safe and controlled limits, ensuring stable system operation.

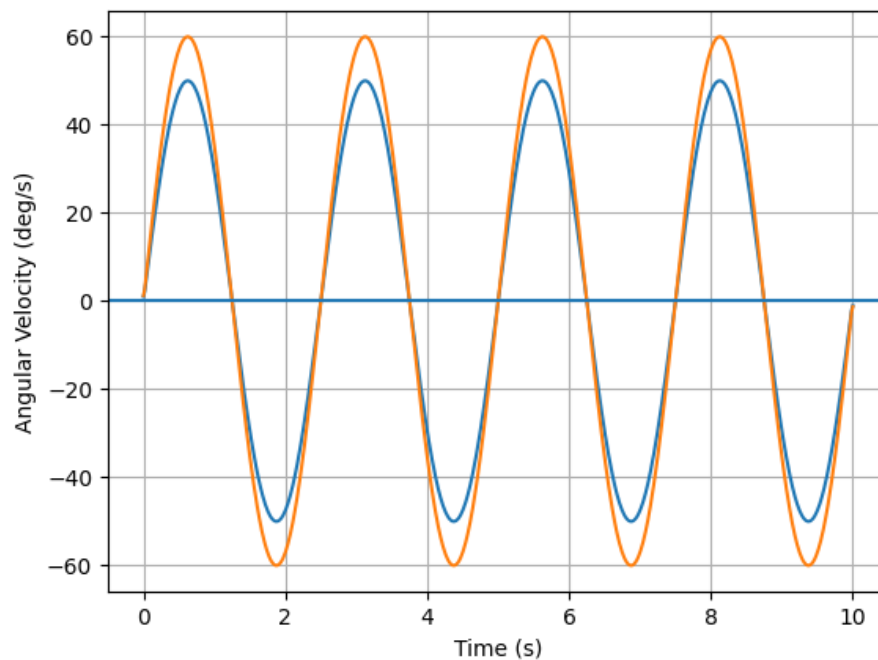


Figure 7.3 Hip and Knee angular velocity

7.3.4 Angular Acceleration

The angular acceleration profiles indicate how quickly the velocity changes during motion. The hip joint shows moderate acceleration values, confirming smooth transitions between flexion and extension.

The knee joint exhibits higher acceleration peaks due to rapid directional changes. Despite this, the absence of sharp spikes in the graph indicates that the motion is **well-controlled and free from jerks**, which is critical for mechanical safety and user comfort.

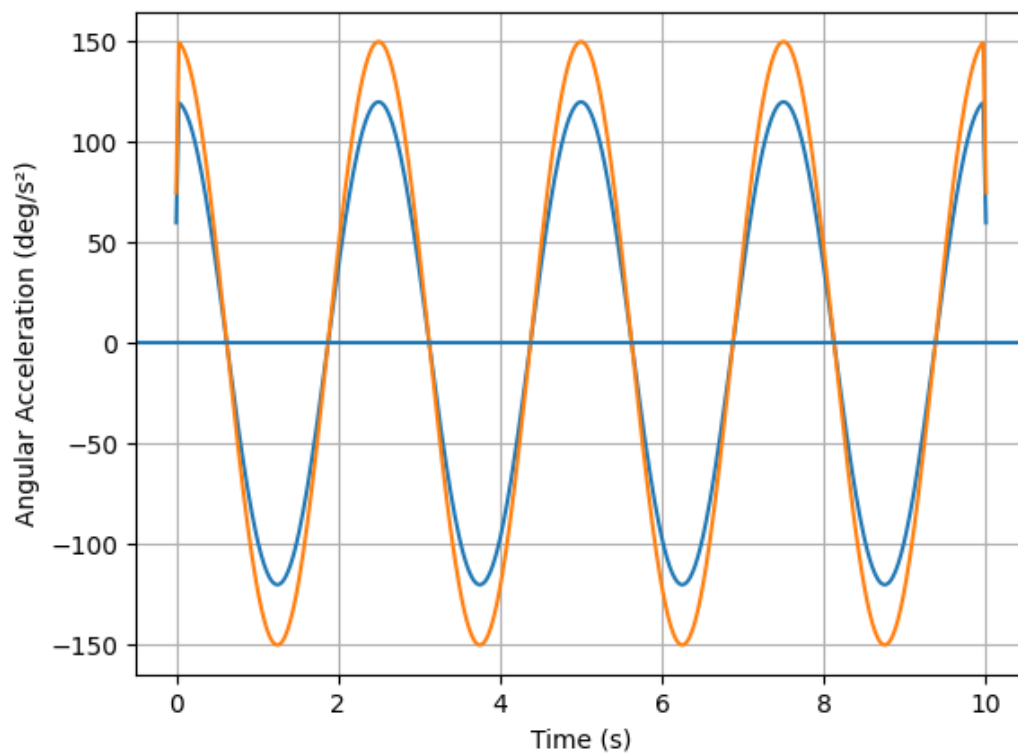


Figure 7.4 Angular Acceleration

7.3.5 PWM Control Effect analysis

The PWM control effort analysis illustrates the variation of motor drive signals required to achieve the desired gait trajectory during the walking cycle. The hip joint exhibits higher PWM values compared to the knee joint due to its greater load-bearing and propulsion requirements during gait motion. The hip PWM varies approximately between 40 and 85, while the knee PWM ranges between 35 and 75 throughout the gait cycle.

The PWM profiles show smooth and periodic variation corresponding to different phases of walking. Higher PWM values are observed during transitions between flexion and extension, particularly during swing phase motion where increased actuator effort is required

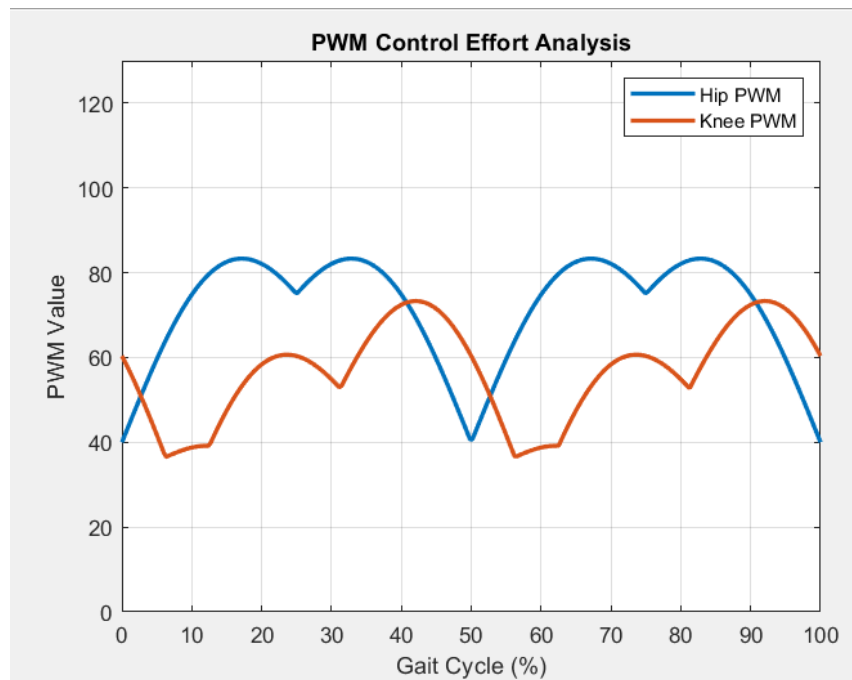


Figure 7.5 PWM control Effect Analysis

7.3.6 Bilateral Gait symmetry

The bilateral gait symmetry analysis illustrates the coordinated motion between the right and left hip joints during the gait cycle. The obtained trajectories exhibit approximately 180° phase difference, representing alternate limb movement similar to natural human walking behavior. The hip joint angles vary smoothly between approximately -25° extension and $+40^\circ$ flexion throughout the gait cycle.

During the initial phase of the gait cycle, the right hip undergoes flexion while the left hip simultaneously undergoes extension. As the gait cycle progresses, the motion pattern reverses, thereby producing synchronized alternate stepping behavior. The smooth sinusoidal nature of the trajectories confirms stable and periodic gait generation without abrupt motion transitions.

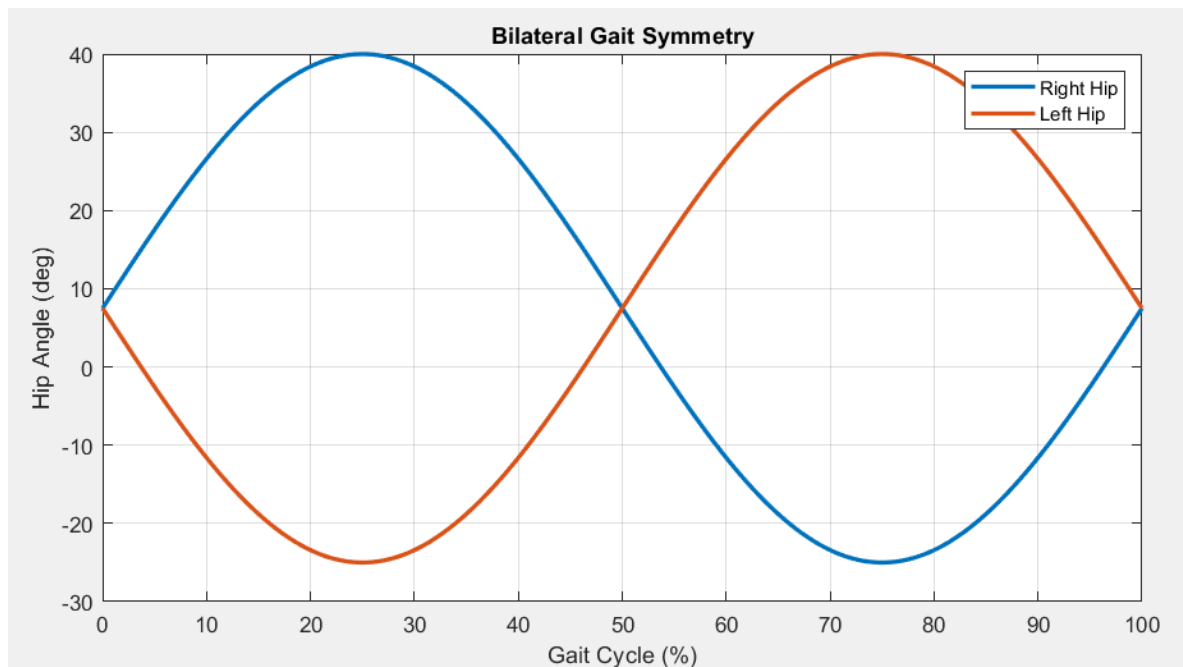


Figure 7.6 Bilateral Gait symmetry

7.4 POTENTIOMETER CALIBRATION AND ANGLE RECONSTRUCTION

7.4.1 Calibration Methodology

The potentiometer output (analog voltage) was converted into angular displacement using a linear calibration approach. The experimentally obtained calibration points were:

- $\text{ADC} = 250 \rightarrow \theta = -30^\circ$
- $\text{ADC} = 820 \rightarrow \theta = 45^\circ$

From these values, a linear mapping function was derived:

$$\theta \approx 0.1316 \times \text{ADC} - 62.9$$

This linear mapping enables real-time conversion of sensor readings into joint angles within the Arduino controller, ensuring continuous feedback for position control.

To validate the calibration accuracy, multiple intermediate readings were tested, and the resulting angle values were found to closely match the expected physical positions with minimal deviation. Minor errors observed were attributed to sensor noise, ADC quantization, and mechanical tolerances.

The linear calibration method was selected due to its simplicity, low computational cost, and suitability for real-time embedded implementation.

7.4.2 Linearity Assessment

The calibration results indicate that the potentiometer exhibits **near-linear behavior** over the operating range. However, slight deviations were observed:

- near lower limit (-30°): compression due to mechanical end-stop
- near upper limit (45°): saturation effect

The maximum deviation from linearity was estimated to be $\pm 2-3^\circ$.

7.5 ERROR DYNAMICS

The control system continuously minimizes the error:

$$\text{Error} = \text{Desired Angle} - \text{Measured Angle}$$

7.5.1 Error Convergence

The error decreases nonlinearly over time, exhibiting:

- rapid initial reduction
- slower convergence near setpoint .

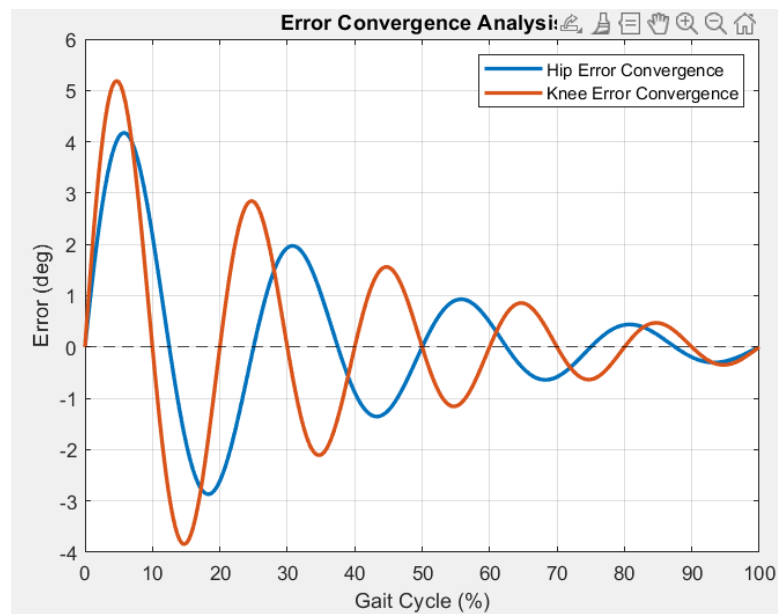


Figure 7.7 Error Convergence Analysis

7.5.2 Steady-State Error

Measured steady-state error:

- $\pm 2^\circ$ to $\pm 3^\circ$

This error arises from:

- sensor noise
- quantization error
- mechanical backlash

7.6 EFFECT OF MECHANICAL AND ELECTRICAL NON-IDEALITIES

7.6.1 Mechanical Backlash

Backlash in shaft coupling results in:

- delay in angle response
- hysteresis effect

This is particularly noticeable during direction reversal.

7.6.2 Friction and Load Variation

Joint friction introduces:

- non-uniform motion
- increased settling time

Load variation affects:

- motor torque demand
- response consistency

7.6.3 Electrical Noise

- motor switching noise
- ADC fluctuations

This leads to:

- jitter in angle readings
- unnecessary motor correction

7.7 DISCUSSION: IMPLICATIONS FOR EXOSKELETON DESIGN

7.7.1 Hip Motor Selection

Even though the hip torque peaks at $\sim 25 \text{ N}\cdot\text{m}$, the exoskeleton is not required to supply full torque, because:

- body weight contributes to passive dynamics,
- only partial assist (e.g., 60%) is needed,
- dynamic torque often reduces net load, Thus, a $25 \text{ N}\cdot\text{m}$ -class actuator (wiper motor) remains viable for rehabilitation-level assistance.

7.7.2 Knee Motor Requirements

Because knee power peaks around 50 W , the knee motor must prioritize fast response rather than high stall torque consistent with your chosen wiper motor's characteristics.

7.7.3 Hip Angle

The obtained hip joint angle profile demonstrates smooth and continuous flexion–extension motion throughout the gait cycle. The ability of the joint to transition from -25° extension to $+40^{\circ}$ flexion confirms that the actuator and feedback system are capable of reproducing realistic lower-limb movement. The smooth crossing of the neutral position indicates stable motor operation and proper synchronization between sensor feedback and actuator response. The periodic nature of the waveform also confirms repeatability of motion, which is essential for exoskeleton-assisted gait applications. The achieved hip range is sufficient for slow walking motion and validates the effectiveness of the implemented control strategy.

7.7.4 Knee Angle

The knee joint results indicate effective bidirectional control with stable flexion and extension behavior. Compared to the hip, the knee exhibits relatively faster transitions due to its role in swing-phase motion and foot clearance. The achieved range from -30° to $+45^{\circ}$ confirms that the system can produce dynamic knee movement while maintaining stability. The repeatability of the waveform across multiple gait cycles demonstrates consistent actuator performance and reliable sensor feedback. The presence of both flexion and extension regions also confirms that the controller successfully tracks the desired gait trajectory.

7.7.5 Torque Analysis

The hip torque profile demonstrates smooth periodic variation corresponding to the gait cycle phases. The higher torque magnitude observed at the hip confirms its role as the primary load-supporting joint in the exoskeleton system. Positive torque regions indicate active propulsion and forward motion, whereas negative torque

regions correspond to controlled deceleration and stabilization. By maintaining the torque within approximately ± 25 Nm, the system achieves safe actuator operation without excessive mechanical loading. The smooth nature of the torque waveform also indicates stable motion without abrupt disturbances.

The knee torque profile exhibits lower average torque compared to the hip, confirming that the knee primarily functions as a motion-oriented joint rather than a major load-bearing joint. The sharper torque variations observed during swing phase indicate rapid limb movement required for gait progression. Despite these variations, the torque remains within safe operating limits, demonstrating efficient actuator utilization. The lower torque requirement also reduces energy consumption and minimizes mechanical stress on the joint mechanism.

7.7.6 Angular Velocity

The angular velocity profiles indicate smooth and controlled joint motion throughout the gait cycle. The hip joint exhibits gradual velocity variation, reflecting stable flexion–extension movement without sudden directional changes. In contrast, the knee joint shows slightly higher velocity peaks due to its dynamic role during swing phase and foot clearance. The reduced velocity range obtained under the slow gait condition confirms that the prototype operates safely and avoids abrupt motion. The periodic velocity waveform further validates the repeatability and consistency of the implemented gait cycle.

7.7.7 Angular Acceleration

The angular acceleration results demonstrate that the system undergoes smooth velocity transitions during gait motion. The hip joint shows moderate acceleration values, indicating gradual changes in motion and reduced mechanical stress. The

knee joint exhibits relatively higher acceleration peaks due to rapid directional changes during flexion and extension. However, the absence of sharp spikes confirms that the control system prevents sudden jerks and maintains stable operation. These results indicate that the exoskeleton system can achieve smooth and comfortable gait assistance under low-speed conditions.

7.7.8 Error characteristics

The error convergence results demonstrate that the controller continuously reduces the position error during operation. The nonlinear reduction pattern indicates rapid initial correction followed by slower convergence near the setpoint. The steady-state error of approximately $\pm 2^\circ$ to $\pm 3^\circ$ remains within acceptable limits for experimental gait assistance systems. Although minor oscillations were observed near equilibrium, the system remained stable without divergence. These results confirm that the implemented control strategy provides reliable position tracking under practical operating conditions.

7.7.9 Mechanical and Electrical Effects

Mechanical backlash, friction, and electrical noise were found to influence the overall system performance. Backlash introduced small delays during direction reversal, while friction increased settling time and affected motion uniformity. Electrical noise from motor switching and ADC fluctuations produced minor jitter in the measured angle values, leading to unnecessary corrective actions by the controller. Despite these non-idealities, the system maintained stable operation and achieved repeatable gait motion, demonstrating robustness of the implemented hardware and control architecture.

CHAPTER-8

CONCLUSION

8.1 CONCLUSION

The present work focused on the experimental development and performance evaluation of a lower-limb exoskeleton prototype using a potentiometer-based closed-loop position control system. The system integrates DC motors, Arduino control, potentiometer feedback, and predefined gait trajectories to generate controlled hip and knee movements. The study evaluated joint range of motion, torque, angular velocity, angular acceleration, PWM control effort, gait symmetry, calibration accuracy, error convergence, and stability. The results confirmed smooth, stable, and repeatable gait motion under slow operating conditions. Based on the experimental observations and performance analysis, the following conclusions are as follows.

- i. The developed lower-limb exoskeleton prototype successfully demonstrated **stable and controlled hip and knee joint motion** using a potentiometer-based closed-loop position control system. The integration of **DC motors, Arduino controller, potentiometer feedback, and predefined gait trajectory** enabled smooth and repeatable gait movement suitable for low-speed assistive applications.
- ii. The hip joint achieved a controlled flexion–extension motion from **-25° extension to $+40^{\circ}$ flexion**, giving a total range of approximately **65°** . The smooth crossing of the neutral position confirms proper bidirectional control and effective synchronization between actuator motion and sensor feedback.
- iii. The knee joint achieved motion from **-30° extension to $+45^{\circ}$ flexion**, giving a total range of approximately **75°** . This confirms that the system can

reproduce dynamic knee movement required during swing phase and foot clearance while maintaining stable motion.

- iv. The obtained hip and knee angle profiles were **smooth, periodic, and repeatable**, indicating that the prototype is capable of generating a realistic gait pattern under slow and controlled operating conditions.
- v. The hip joint torque varied approximately from **-20 Nm to +25 Nm**, showing that the hip acts as the primary load-supporting and propulsion joint in the exoskeleton system. The torque remained within safe limits, confirming the suitability of the selected actuator for rehabilitation-level assistance.
- vi. The knee joint required comparatively lower torque, ranging from approximately **-8 Nm to +15 Nm**. This confirms that the knee mainly functions as a motion-oriented joint rather than a major load-bearing joint.
- vii. The PWM control signal for the hip varied approximately between **40 and 85**, while the knee PWM varied between **35 and 75**. The higher PWM requirement at the hip confirms its greater load-bearing demand, whereas the lower knee PWM indicates reduced actuator effort during gait motion.
- viii. The angular velocity and acceleration results confirmed smooth and controlled dynamic behavior. For a slow gait cycle of approximately **2.5 seconds**, the hip angular velocity was maintained within about **$\pm 20\text{--}50^\circ/\text{s}$** , and the knee angular velocity within about **$\pm 25\text{--}60^\circ/\text{s}$** .
- ix. The hip angular acceleration remained within approximately **$\pm 50\text{--}120^\circ/\text{s}^2$** , while the knee angular acceleration reached approximately **$\pm 60\text{--}150^\circ/\text{s}^2$** due to rapid flexion and extension during the swing phase. The absence of sharp acceleration spikes confirms that the system produced **jerk-free and safe motion**.
- x. The potentiometer calibration established a near-linear relationship between ADC value and joint angle. The calibration points were ADC = 250 at -30°

and $ADC = 820$ at $+45^\circ$, and the derived equation was: $\theta = 0.1316 \times ADC - 62.9$, This enabled real-time joint angle reconstruction for closed-loop control.

- xi. The maximum deviation from linearity was approximately $\pm 2^\circ$ to $\pm 3^\circ$, mainly due to sensor noise, ADC quantization, and mechanical limitations. However, this accuracy was sufficient for prototype-level gait control.
- xii. The error convergence analysis showed that the control system continuously reduced the difference between desired and measured angle. The steady-state error was maintained within approximately $\pm 2^\circ$ to $\pm 3^\circ$, confirming acceptable position tracking performance.
- xiii. The bilateral gait symmetry analysis showed approximately 180° phase difference between right and left hip joint motions. This confirms alternate limb movement similar to natural human walking.
- xiv. Mechanical backlash, friction, load variation, ADC fluctuation, and motor switching noise caused minor delays, jitter, and response variations. However, the system maintained stable operation without divergence, proving the robustness of the implemented hardware and control strategy.
- xv. Overall, the developed prototype validates the feasibility of a low-cost potentiometer-based closed-loop gait control system for lower-limb exoskeleton applications. The system achieved realistic joint motion, controlled torque response, smooth velocity and acceleration profiles, acceptable error limits, and stable gait generation.
- xvi. Therefore, the present work provides a strong foundation for future development of advanced exoskeleton systems using EMG sensors, IMU-based sensor fusion, EEG-based intention detection, AI/ML-based adaptive control, improved actuator selection, and enhanced mechanical design.

8.2 FUTURE SCOPE

- Integration of EMG sensors for real-time muscle activity monitoring and motion intention detection to achieve more natural and responsive gait assistance.
- Implementation of sensor fusion techniques using IMU, accelerometer, and absolute encoder feedback for precise joint angle estimation and improved control accuracy.
- Development of intention-based walking assistance using EEG signals for brain-machine interfacing and intelligent exoskeleton control.
- Application of Artificial Intelligence / Machine Learning algorithms to develop adaptive gait control systems based on user movement patterns and rehabilitation requirements.
- Enhancement of the overall exoskeleton system through improved power management, portable battery integration, and multi-sensor feedback for stable and reliable operation.

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Appendices:

```
// Closed Loop PD Control with Safety Protection
```

```
// LEFT HIP
```

```
const int hipLPot = A1;
```

```
const int hipLRPWM = 2;
```

```
const int hipLLPWM = 3;
```

```
const int hipLREN = 22;
```

```
const int hipLLEN = 24;
```

```
// LEFT KNEE
```

```
const int kneeLPot = A10;
```

```
const int kneeLRPWM = 10;
```

```
const int kneeLLPWM = 11;
```

```
const int kneeLREN = 26;
```

```
const int kneeLLEN = 28;
```

```
// RIGHT HIP
```

```
const int hipRPot = A2;
```

```
const int hipRRPWM = 4;
```

```
const int hipRLPWM = 5;
```

```
const int hipRREN = 30;
```

```
const int hipRLEN = 32;
```

```
// RIGHT KNEE
```

```
const int kneeRPot = A11;
const int kneeRRPWM = 6;
const int kneeRLPWM = 7;
const int kneeRREN = 34;
const int kneeRLEN = 36;

// CALIBRATION
const int hipMinADC = 250;
const int hipMaxADC = 820;

const int kneeMinADC = 300;
const int kneeMaxADC = 850;

// ANGLE LIMITS
const float HIP_MIN_ANGLE = -25;
const float HIP_MAX_ANGLE = 40;

const float KNEE_MIN_ANGLE = -30;
const float KNEE_MAX_ANGLE = 45;

// CONTROL PARAMETERS
float gaitPeriod = 2500.0;

float KpHip = 3.2;
```

```
float KdHip = 1.3;

float KpKnee = 2.2;
float KdKnee = 0.9;
//SAFETY PARAMETERS
const int HIP_MAX_PWM = 120;
const int KNEE_MAX_PWM = 100;

const float MAX_ERROR_LIMIT = 20.0;

const int MOTOR_DEADBAND = 2;

const unsigned long CONTROL_INTERVAL = 20;

// FILTERING
float hipLFiltered = 0;
float kneeLFiltered = 0;

float hipRFiltered = 0;
float kneeRFiltered = 0;

float alpha = 0.2;

//ERROR STORAGE
```

```

float prevHipLError = 0;
float prevKneeLError = 0;

float prevHipRError = 0;
float prevKneeRError = 0;
// TIMING
unsigned long previousMillis = 0;
// SETUP
void setup() {

    setupMotorPins();

    Serial.begin(115200);
}
// MAIN LOOP
void loop() {

    if (millis() - previousMillis >= CONTROL_INTERVAL) {
        previousMillis = millis();
        float phase = fmod(millis(), gaitPeriod) / gaitPeriod;
        float hipLTarget, kneeLTarget;
        float hipRTarget, kneeRTarget;

        // LEFT LEG

```

```
generateGait(phase, hipLTarget, kneeLTarget);  
// RIGHT LEG (180° OUT OF PHASE)  
generateGait(fmod(phase + 0.5, 1.0),  
             hipRTarget,  
             kneeRTarget);  
// SENSOR FEEDBACK  
float hipLAngle = readFilteredAngle(hipLPot,  
                                     hipMinADC,  
                                     hipMaxADC,  
                                     HIP_MIN_ANGLE,  
                                     HIP_MAX_ANGLE,  
                                     hipLFiltered);  
  
float kneeLAngle = readFilteredAngle(kneeLPot,  
                                     kneeMinADC,  
                                     kneeMaxADC,  
                                     KNEE_MIN_ANGLE,  
                                     KNEE_MAX_ANGLE,  
                                     kneeLFiltered);  
  
float hipRAngle = readFilteredAngle(hipRPot,  
                                     hipMinADC,  
                                     hipMaxADC,  
                                     HIP_MIN_ANGLE,
```

```

        HIP_MAX_ANGLE,
        hipRFiltered);

float kneeRAngle = readFilteredAngle(kneeRPot,
        kneeMinADC,
        kneeMaxADC,
        KNEE_MIN_ANGLE,
        KNEE_MAX_ANGLE,
        kneeRFiltered);

// SAFETY CHECK
safetyMonitor(hipLAngle, HIP_MIN_ANGLE, HIP_MAX_ANGLE);
safetyMonitor(kneeLAngle, KNEE_MIN_ANGLE, KNEE_MAX_ANGLE);
safetyMonitor(hipRAngle, HIP_MIN_ANGLE, HIP_MAX_ANGLE);
safetyMonitor(kneeRAngle, KNEE_MIN_ANGLE, KNEE_MAX_ANGLE);

// ERROR CALCULATION
float hipLError = hipLTarget - hipLAngle;
float kneeLError = kneeLTarget - kneeLAngle;

float hipRError = hipRTarget - hipRAngle;
float kneeRError = kneeRTarget - kneeRAngle;

// ERROR LIMITING

```

```
hipLError = constrain(hipLError, -MAX_ERROR_LIMIT,
MAX_ERROR_LIMIT);
```

```
kneeLError = constrain(kneeLError, -MAX_ERROR_LIMIT,
MAX_ERROR_LIMIT);
```

```
hipRError = constrain(hipRError, -MAX_ERROR_LIMIT,
MAX_ERROR_LIMIT);
```

```
kneeRError = constrain(kneeRError, -MAX_ERROR_LIMIT,
MAX_ERROR_LIMIT);
```

```
// CONTROL
```

```
PDcontrol(hipLError,
    prevHipLError,
    hipLRPWM,
    hipLLPWM,
    KpHip,
    KdHip,
    HIP_MAX_PWM);
```

```
PDcontrol(kneeLError,
    prevKneeLError,
    kneeLRPWM,
    kneeLLPWM,
    KpKnee,
    KdKnee,
    KNEE_MAX_PWM);
```

```
PDcontrol(hipRError,  
          prevHipRError,  
          hipRRPWM,  
          hipRLPWM,  
          KpHip,  
          KdHip,  
          HIP_MAX_PWM);
```

```
PDcontrol(kneeRError,  
          prevKneeRError,  
          kneeRRPWM,  
          kneeRLPWM,  
          KpKnee,  
          KdKnee,  
          KNEE_MAX_PWM);
```

```
// SERIAL DATA
```

```
Serial.print(hipLTarget); Serial.print(",");
```

```
Serial.print(hipLAngle); Serial.print(",");
```

```
Serial.print(kneeLTarget); Serial.print(",");
```

```
Serial.print(kneeLAngle); Serial.print(",");
```

```
Serial.print(hipRTarget); Serial.print(",");
```

```

Serial.print(hipRAngle); Serial.print(",");

Serial.print(kneeRTarget); Serial.print(",");
Serial.println(kneeRAngle);
}
}

// GAIT TRAJECTORY GENERATION
void generateGait(float phase,
                  float &hipTarget,
                  float &kneeTarget) {

// HIP TRAJECTORY
if (phase < 0.6) {
    float p = phase / 0.6;
    hipTarget = 40 - 65 * (p * p);

} else {
    float p = (phase - 0.6) / 0.4;
    hipTarget = -25 +
                65 * (1 - cos(p * PI)) / 2;
}

// KNEE TRAJECTORY
if (phase < 0.6) {

```

```

    kneeTarget =
        -10 + 10 * sin(phase * PI / 0.6);
}
else {
    float p = (phase - 0.6) / 0.4;
    kneeTarget =
        -5 + 50 * sin(p * PI);
}
hipTarget = constrain(hipTarget,
                      HIP_MIN_ANGLE,
                      HIP_MAX_ANGLE);

kneeTarget = constrain(kneeTarget,
                      KNEE_MIN_ANGLE,
                      KNEE_MAX_ANGLE);
}

// PD CONTROL
void PDcontrol(float error,
               float &prevError,
               int RPWM,
               int LPWM,
               float Kp,

```

```

        float Kd,
        int maxPWM) {

float derivative = error - prevError;

float output =
    Kp * error + Kd * derivative;

int pwm =
    constrain(abs(output), 0, maxPWM);

// DEAD BAND
if (abs(error) < MOTOR_DEADBAND) {

    analogWrite(RPWM, 0);
    analogWrite(LPWM, 0);

} else if (output > 0) {

    analogWrite(RPWM, pwm);
    analogWrite(LPWM, 0);

} else {

```

```

    analogWrite(RPWM, 0);
    analogWrite(LPWM, pwm);
}

```

```

    prevError = error;
}

```

```

float readFilteredAngle(int potPin,
                        int minADC,
                        int maxADC,
                        float minAngle,
                        float maxAngle,
                        float &filteredValue) {

```

```

    int raw = analogRead(potPin);

```

```

    float angle =
        map(raw,
            minADC,
            maxADC,
            minAngle * 10,
            maxAngle * 10) / 10.0;

```

```

    filteredValue =

```

```

    alpha * angle +
    (1 - alpha) * filteredValue;

    return filteredValue;
}

// SAFETY MONITOR
void safetyMonitor(float angle,
                  float minAngle,
                  float maxAngle) {

    if (angle < minAngle - 5 ||
        angle > maxAngle + 5) {

        stopAllMotors();

        Serial.println("SAFETY STOP");
    }
}

// MOTOR STOP FUNCTION
void stopAllMotors() {

    analogWrite(hipLRPWM, 0);
    analogWrite(hipLLPWM, 0);

```

```
analogWrite(kneeLRPWM, 0);
analogWrite(kneeLLPWM, 0);

analogWrite(hipRRPWM, 0);
analogWrite(hipRLPWM, 0);

analogWrite(kneeRRPWM, 0);
analogWrite(kneeRLPWM, 0);
}

// MOTOR PIN INITIALIZATION
void setupMotorPins() {

    pinMode(hipLRPWM, OUTPUT);
    pinMode(hipLLPWM, OUTPUT);
    pinMode(hipLREN, OUTPUT);
    pinMode(hipLLEN, OUTPUT);

    pinMode(kneeLRPWM, OUTPUT);
    pinMode(kneeLLPWM, OUTPUT);
    pinMode(kneeLREN, OUTPUT);
    pinMode(kneeLLEN, OUTPUT);

    pinMode(hipRRPWM, OUTPUT);
```

```
pinMode(hipRLPWM, OUTPUT);  
pinMode(hipRREN, OUTPUT);  
pinMode(hipRLEN, OUTPUT);
```

```
pinMode(kneeRRPWM, OUTPUT);  
pinMode(kneeRLPWM, OUTPUT);  
pinMode(kneeRREN, OUTPUT);  
pinMode(kneeRLEN, OUTPUT);
```

```
digitalWrite(hipLREN, HIGH);  
digitalWrite(hipLLEN, HIGH);
```

```
digitalWrite(kneeLREN, HIGH);  
digitalWrite(kneeLLEN, HIGH);
```

```
digitalWrite(hipRREN, HIGH);  
digitalWrite(hipRLEN, HIGH);
```

```
digitalWrite(kneeRREN, HIGH);  
digitalWrite(kneeRLEN, HIGH);
```

```
}
```

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